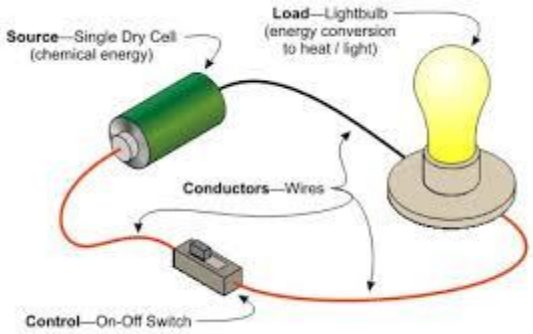
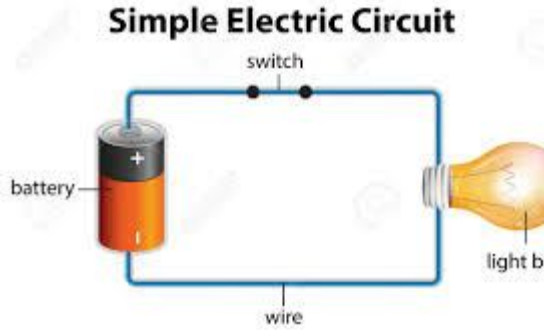
Chapter 4:

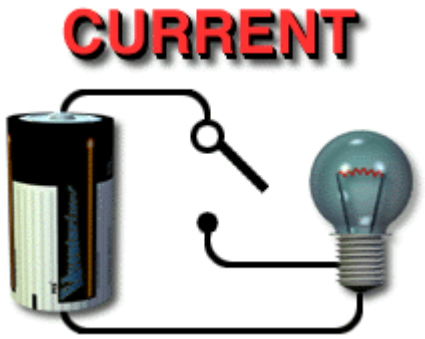
Elements used in Electrical Engineering & Electric Circuits

4.1 Introduction (Electric circuit& Elements)

- it is simply a systematic interconnection of the electrical elements so that it can perform some function.
- They are two types of electrical elements:
  - ❖ Active elements(voltage source ,current source)
  - ❖ Passive elements(resistor, inductor, capacitor)
- An active element is capable of generating energy while passive element is not.





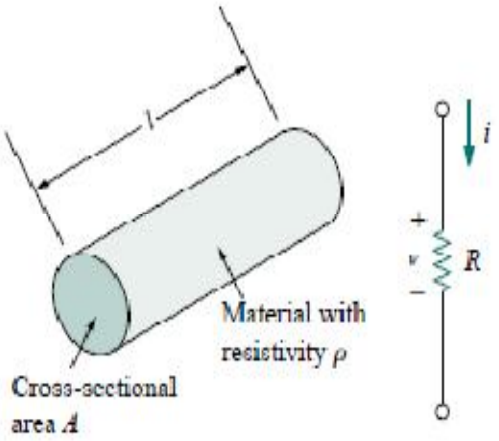


4.2 Elements used in Electrical Engineering
4.2.1.Passive Elements ;4.2.1.1 Resistor



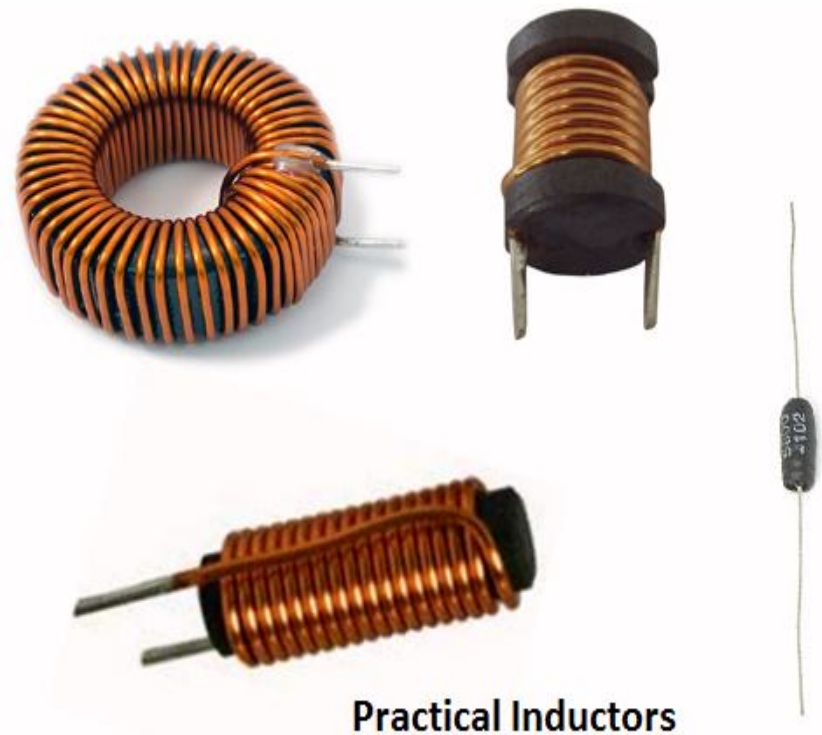
- Materials in general have a characteristic of resisting the flow of electric charge.
- This physical property, or ability to resist current, is known as *resistance* and is represented by the symbol *R*. The resistance of a conductor material with a uniform cross-sectional area *A* depends on *A* and its length *l*.
- In mathematical form,

$R = \rho(l/A)$ where ρ is known as the *resistivity* of the material in ohm-meters

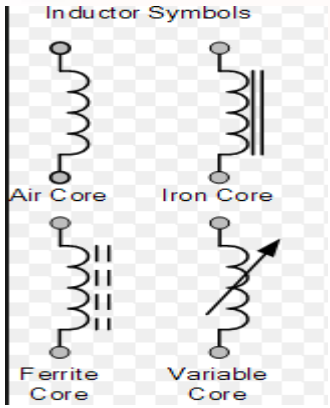


4.2.1.2.Inductor

- An inductor is a passive electrical component that can store energy in a magnetic field created by the electric current passing through it.
- An inductor's ability to store magnetic energy is measured by its inductance, in units of henries



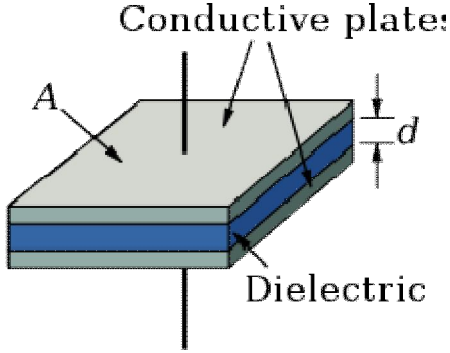
Practical Inductors

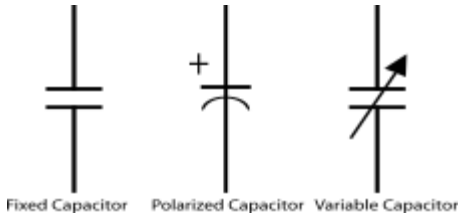


4.2.1.3 Capacitor

- a capacitor is nothing but any conducting surfaces separated by an insulating medium called dielectric (air, paper, mica, etc.)
  - **Capacitance**: it is defined as the amount of charge required to create a unit potential
  - i.e C=Q/V=$\varepsilon A/d$ where ε is called permittivity of the dielectric material







4.2.1.4 Poteniometer

- Poteniometer is a variable resistance
- Found on many household devices
- Rotating disc changes the length of the resistor

4.2.2. Active Elements
4.2.2.1 Voltage source

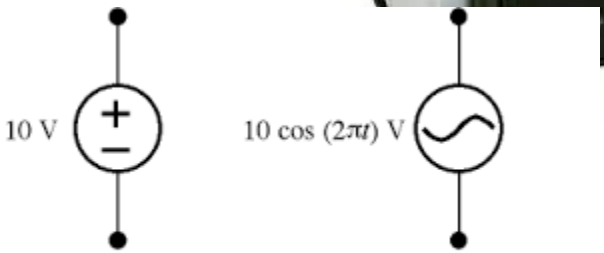
- An ideal independent voltage source is an active element that provides a specified voltage that is completely independent of other circuit variables

Examples: generator, batteries





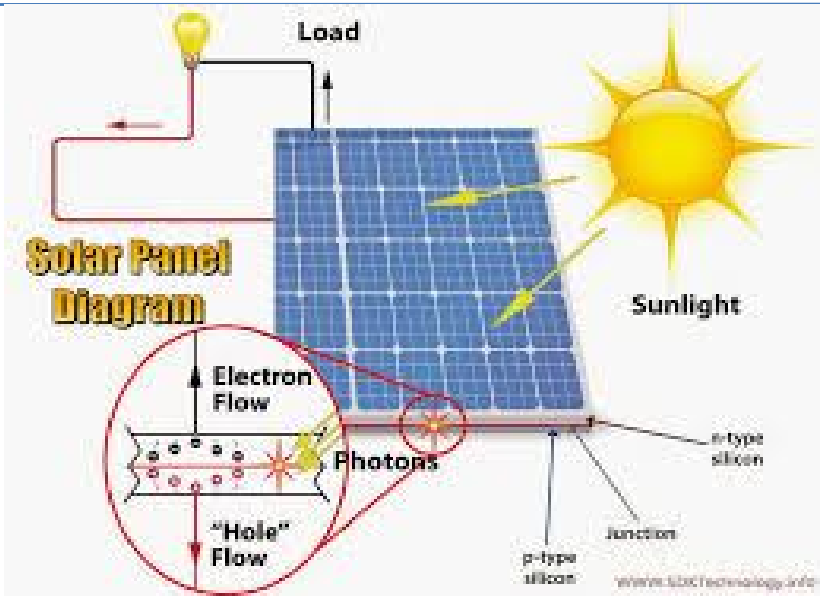


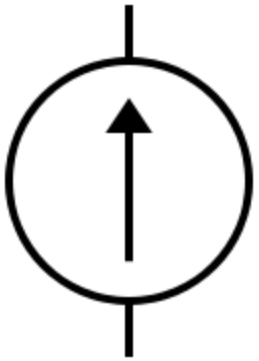


(a) DC voltage source (b) AC voltage source

4.2.2.2 Current source

- An ideal independent current source is an active element that provides a specified current that is completely independent of other circuit variables
- **Examples:** solar PV cells.





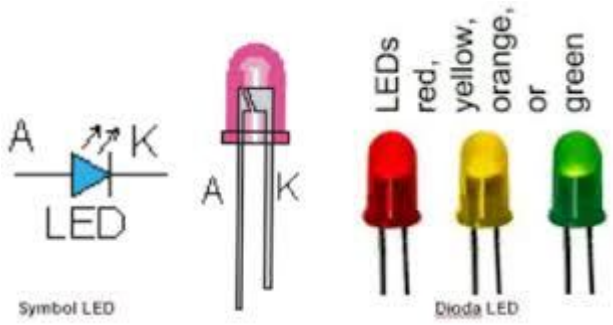
4.2.2.3 Diode

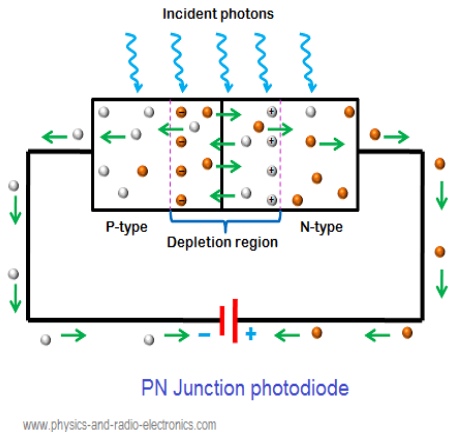
- A diode is a two-terminal device made up of semiconductor materials. Diodes have two active electrodes between which the signal of interest may flow, and most are used for their unidirectional electric current property.
- The directionality of current flow most diodes exhibit is sometimes generically called the rectifying property.
- The most common function of a diode is to allow an electric current to pass in one direction (called the forward biased condition) and to block the current in the opposite direction (the reverse biased condition).



Diode types

- There are several types of junction diodes with different applications: Some of them are:
  - Normal (p-n) diodes(used in rectifiers, clippers, clampers)
  - Light-emitting diodes(indicators, LED bulbs, LED T.V.)
  - Photodiodes(PV cells)
  - Zener diodes (voltage regulator)



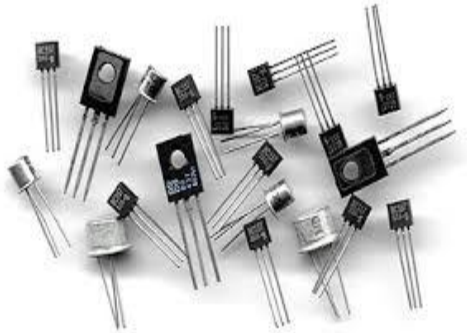


4.2.2.4 Transistor:

- A transistor is a semiconductor device commonly used to **amplify** or **switch** electronic signals.
- A transistor is made of a solid piece of a semiconductor material, with at least three terminals for connection to an external circuit.



- A voltage or current applied to one pair of the transistor's terminals changes the current flowing through another pair of terminals.
- Because the controlled (output) power can be much larger than the controlling (input) power, the transistor provides amplification of a signal.



Transistor

- The transistor is the fundamental building block of modern electronic devices, and is used in radio, telephone, computer and other electronic systems.
- Some transistors are packaged individually but most are found in integrated circuits.
- Transistors are classified into two broad categories. These are:
    - Bipolar Junction Transistors(BJT)
    - Field Effect Transistors (FET)

.....

- Do you believe holiday decoration lights are wired in series or in parallel?

...

- What is the resistance of a 25W light bulb when attached to a 120V outlet?

4.3 Electric Circuits

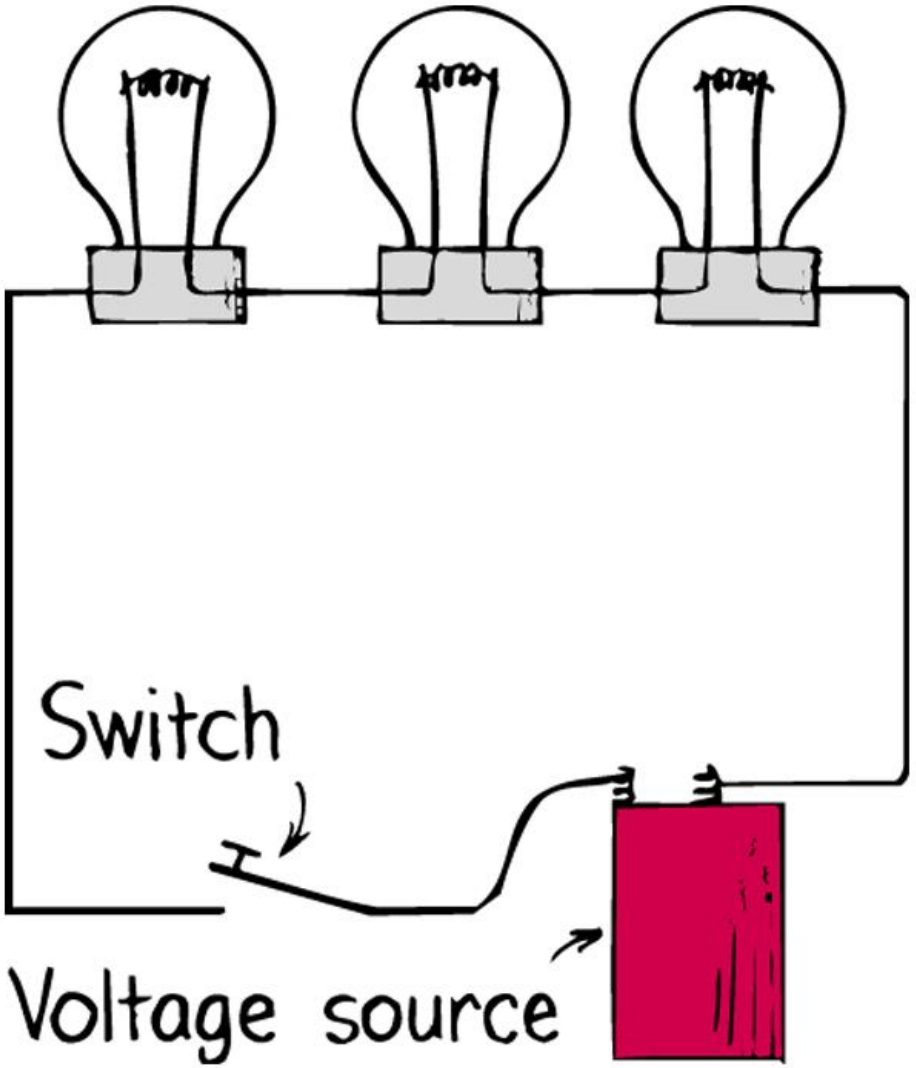
4.3.1 The Basics

- Circuit: Any path along which electrons can flow
  - Electrons will only flow if the circuit is complete with no gaps
- Open Circuits are broken and do not allow the flow of electrons
- Closed Circuits are complete and allow the flow of electrons

4.3.2 Types of Circuits

- Devices within a circuit can be connected in one of two ways:
  - In Series: Devices form a single pathway for electrons to flow
  - In Parallel: Devices form branches, each of which is a separate path for the flow of electrons

4.3.2.1 Series Circuits



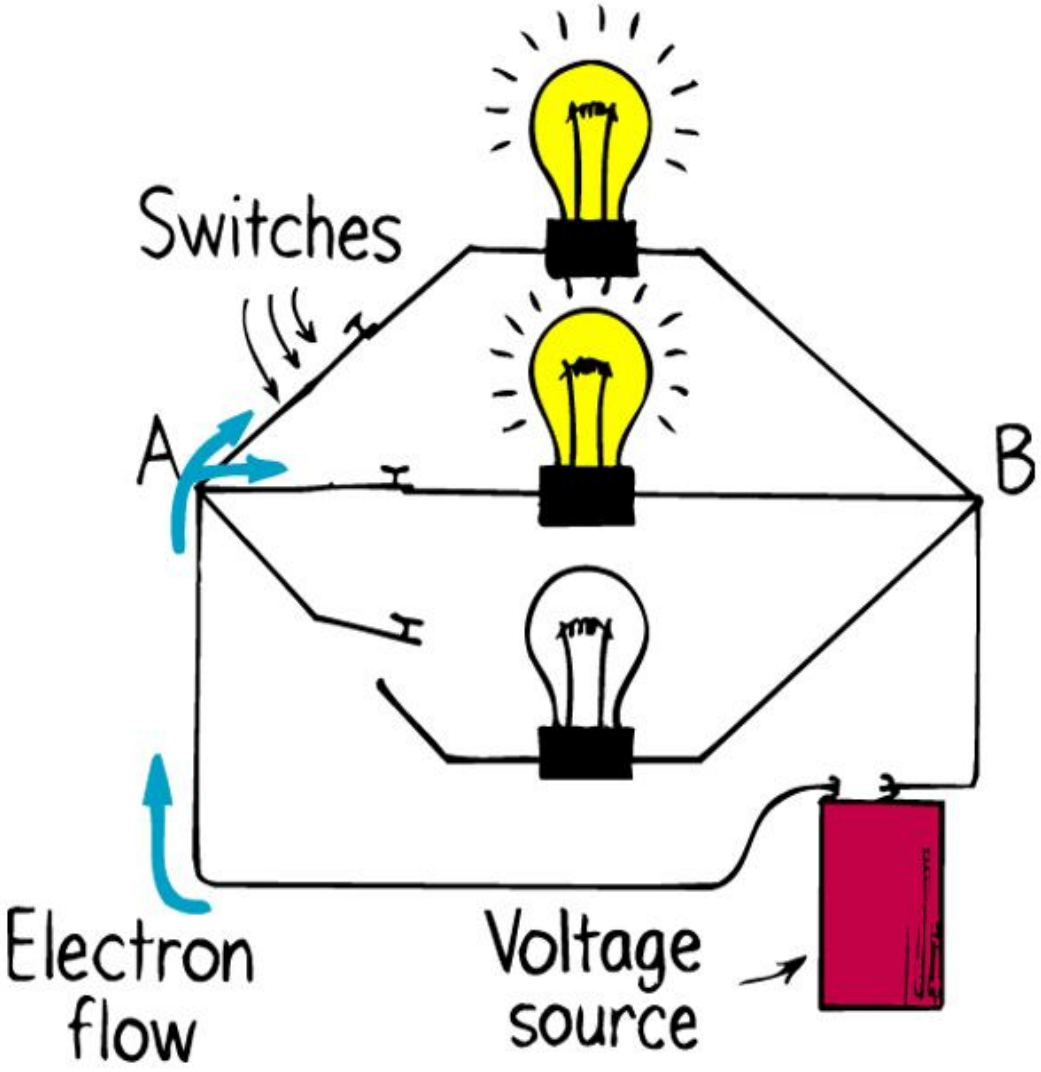
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Series Circuits

- All current passing through each device is the same
- Ohm's Law (I = V/R) applies to the entire circuit, AND each individual device within the circuit

- Drawback: If one device in the circuit fails, the entire circuit will fail
  - i.e.) If one light goes out, they all go out

4.3.2.2 Parallel Circuits



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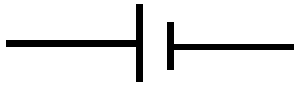
Parallel Circuits

- Each device connects the same 2 points in the circuit
  - Voltage is the same across each branch
- The total current in the circuit is the sum of the currents in each branch
- More parallel branches = less circuit resistance = MORE CURRENT

Parallel Circuits

- Advantage: If one device in the circuit fails, the rest of the circuit still functions
  - i.e.) If one light goes out, the others stay lit

Schematic Diagrams

- Simple diagrams which describe electric circuits
  - Connecting Wire
  -  Battery
  -  Resistor
  -  Capacitor
  -  Open Switch
  -  Closed Switch

Schematic Diagrams - Examples

- Two series 3 Ω resistors in parallel with a 1 μF capacitor, connected to a 12 V battery
- Three parallel 2 μF capacitors connected in series to a 10 Ω resistor and 1.5 V battery
- Commonly used symbols



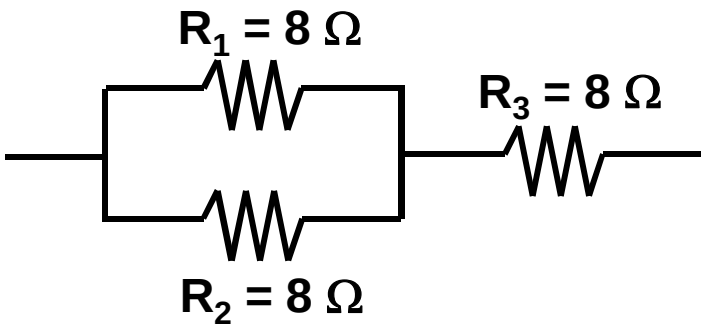
...

- Draw the following Schematics
- Two series resistors in parallel with a capacitor, connected to a battery

- Three parallel capacitors connected in series to a resistor and battery

4.3.3 Equivalent Resistance

- Combining resistors in a compound circuit:
  - Equivalent Resistance: The value of the single resistor that would represent the resistance of all of the devices in a circuit



Equivalent Resistance

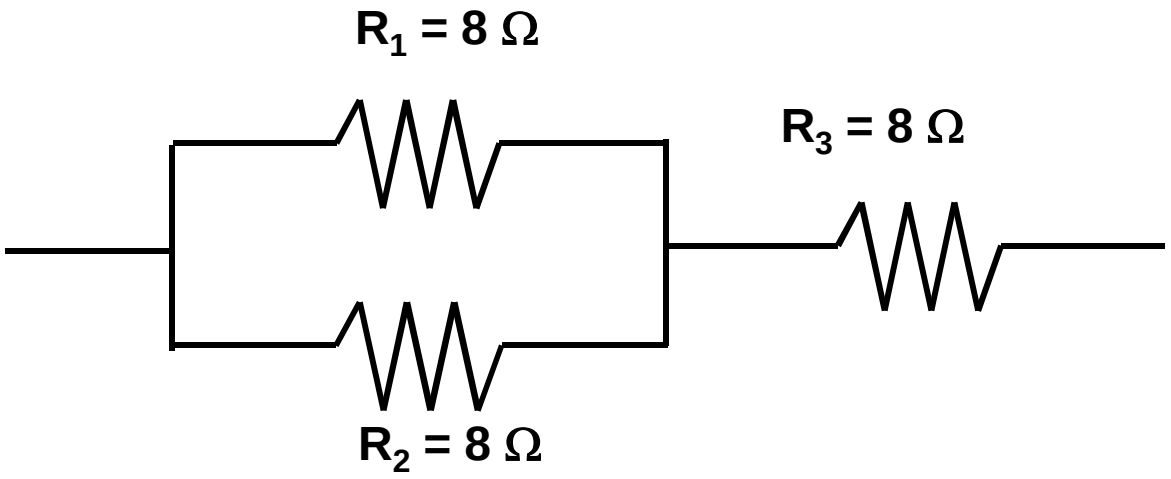
- For series circuits:

$$R_{eq} = R_1 + R_2 + R_3 + \ldots$$

- For parallel circuits:

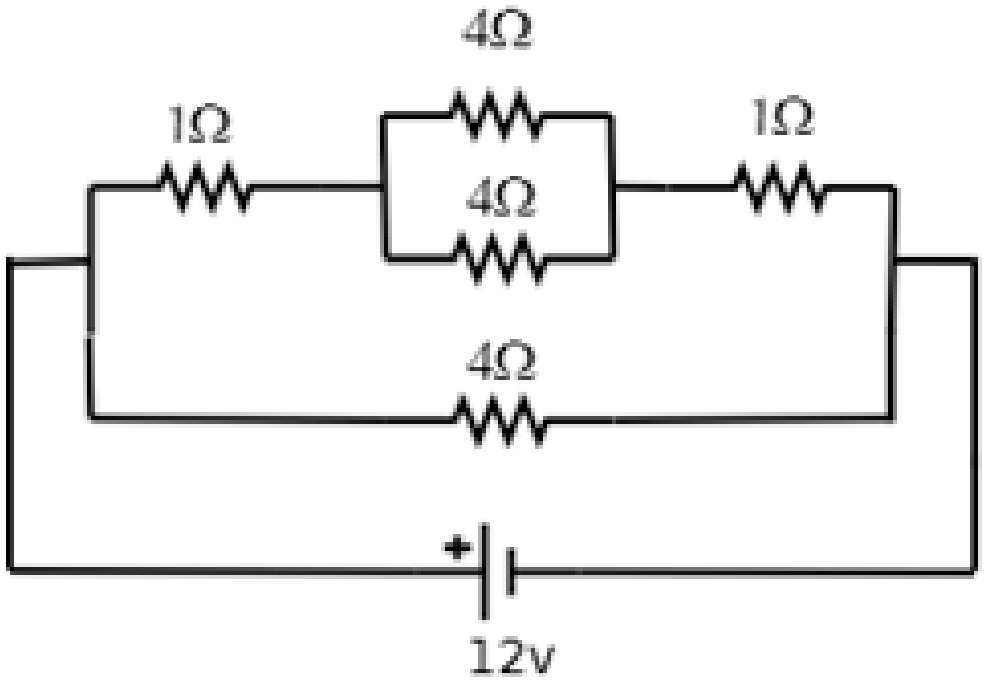
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \ldots$$

- Calculate the Equivalent Resistance for the circuit below…



...

- What is the equivalent resistance for the circuit below?



...

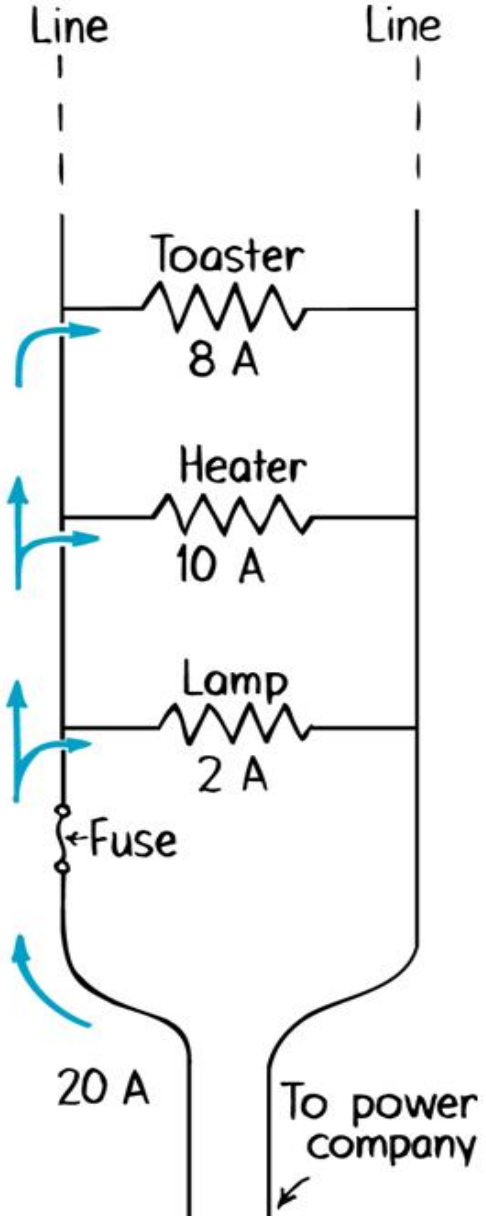
- How much power is dissipated by a 24,000Ω as 120V are sent across it?

4.3.4 Parallel Circuits and Overloading

- Lines: Low resistance wires connecting wall outlets to generators at the power utility
  - Usually carry 110 V to 120 V
- When more devices connected in parallel:
  - Resistance is lowered
  - Current is increased

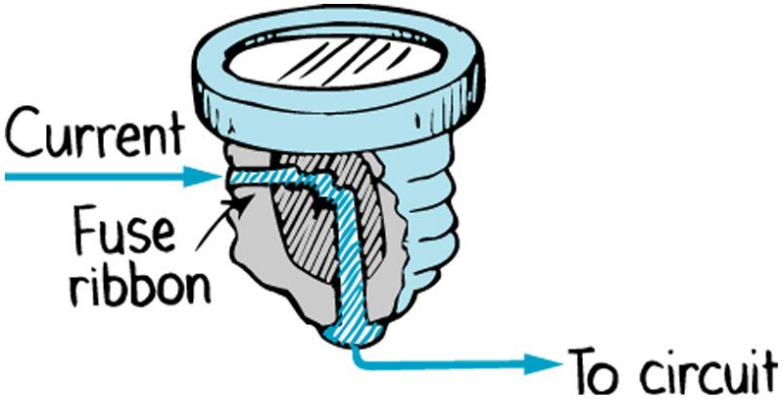
Parallel Circuits and Overloading

- If this current exceeds the safe amount for the line, the circuit is said to be <u>overloaded</u>
  - Resulting heat can melt insulation and cause a fire



Preventing Overloading

- Fuses: Devices connected in series to prevent the overloading of circuits
  - A fuse is constructed of a thin wire that will melt if the current becomes to great, therefore breaking the circuit



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4.3.5 Short Circuits

- Short Circuit: Insulation wears away, causing wires to touch, shortening the circuit
  - This bypasses the normal resistance of the circuit
  - Draws very large currents

Chapter 5:
Electrical Machines

Contents

- Basic construction and principles
- DC machines
- Synchronous machines
- Induction machines

objectives

When you have studied this chapter ,you should ;

- **Have an understanding of electrical machines construction**
- **Understand the principles of DC machines**
- **Understand the principles and application of Synchronous machines**
- **Understand the principles and application of induction machines**

Basic Construction, Introduction

One of energy can be obtained from the other form with the help of converters. Converters that are used to continuously translate electrical input to mechanical output or vice versa are called electric machines. The process of translation is known as electromechanical energy conversion.

An electrical machine is link between an electrical system and a mechanical system.

Conversion from mechanical to electrical: generator

Conversion from electrical to mechanical :motor

Classification

Electrical machines are classified as:-

- **DC machines and AC machines**
- **AC machines further classified as Sychronous machine and Induction machine**
- **Machines are called AC machines (generators or motors) if the electrical system is AC.**
- **DC machines (generators or motors) if the electrical system is DC.**

1.Generator

- A generator is a electromechanical device which converts mechanical energy to electrical energy

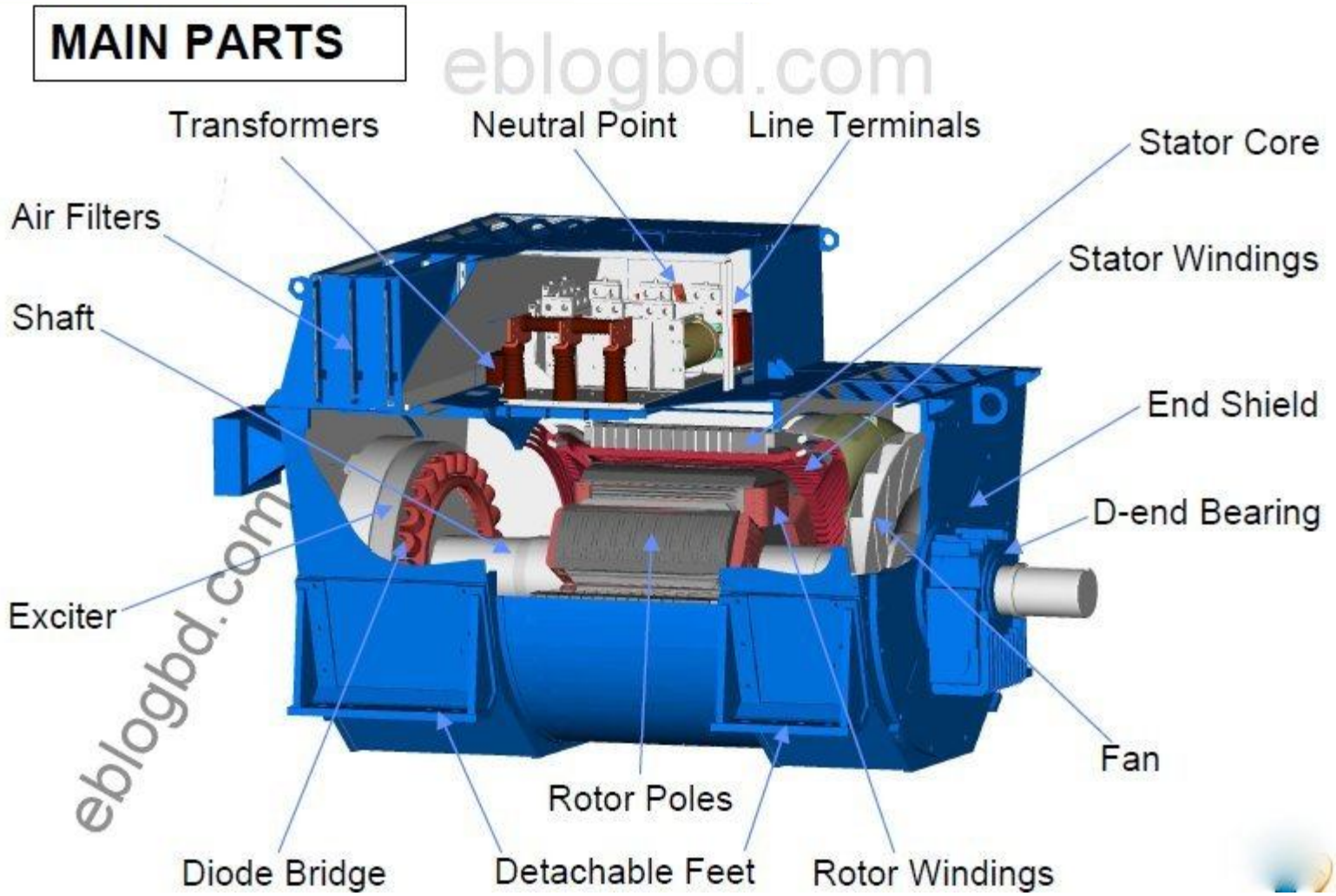
Types of generators:

- AC generators
  - Synchronous generator or alternator(generally used at generating station)
  - Asynchronous generator or induction generator(generally used with wind mills)
- DC generators (obsolete now a days)

Generator

Synchronous Generator

MAIN PARTS



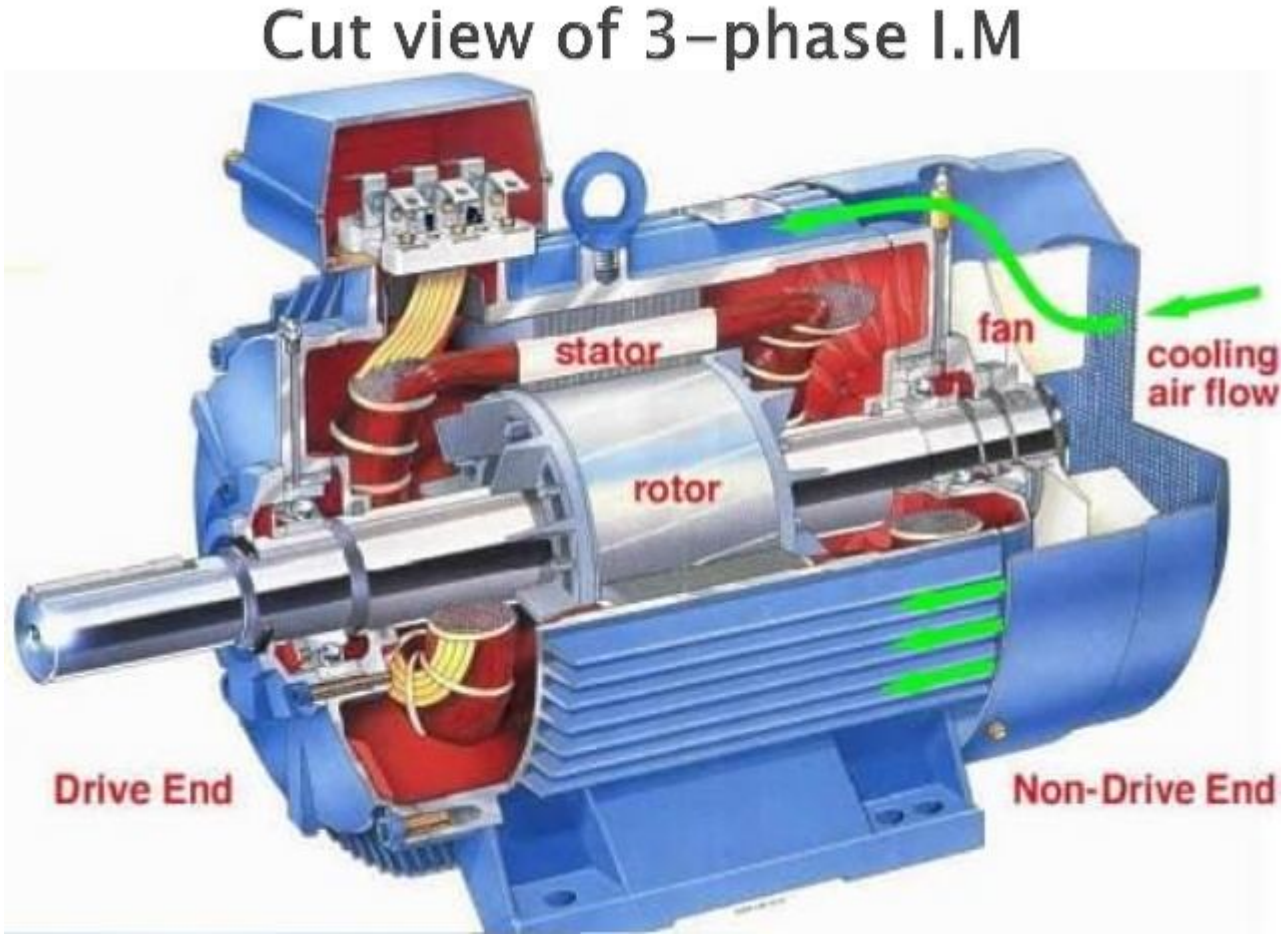
2.Motor

- A motor is a electromechanical device which converts electrical energy into mechanical energy

Type of motors:

- AC motors
  - ❖ Synchronous motor(generally used for constant speed applications)
- Asynchronous motor or induction motor(generally 70 to 80% of the load is induction motor load)
- DC motors(used in food industry ,railways ,textiles etc.)

Motor Continued…

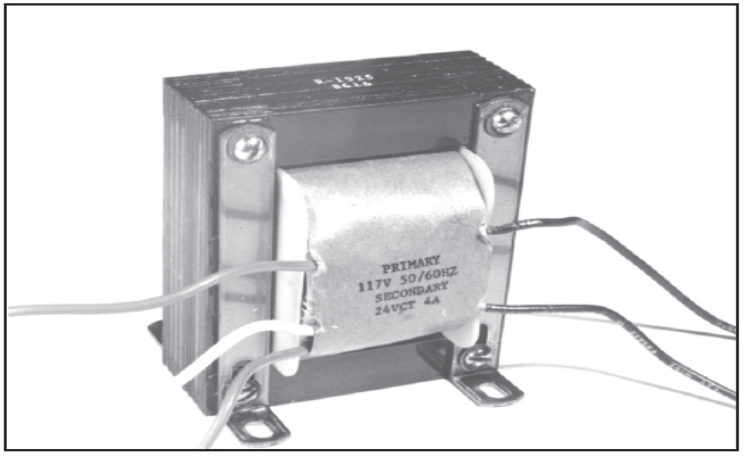


3.Transformers :

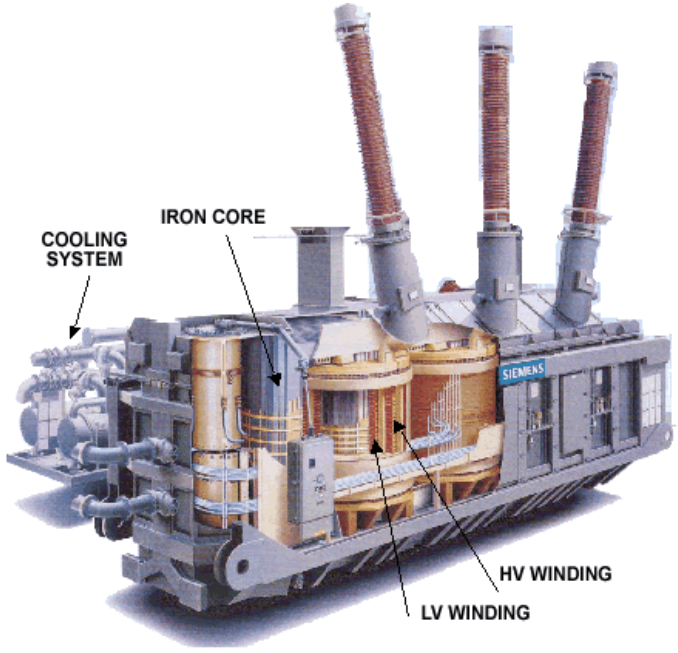
- It is electromagnetic device which transfers electrical energy from one circuit to another circuit in the presence of common magnetic field.
- Transformers are electrical devices used to convert or "transform" AC voltage from one level to another. (high to low or low to high)
- Input and output are AC
- They do this by the principle of electromagnetic induction

Transformer -Types & Applications

- **Types of transformers**
  - Step up transformer
  - Step down transformer
  - Isolation transformer
  - Control transformer
  - Power transformer
  - Distribution transformer
  - *Auto transformer*

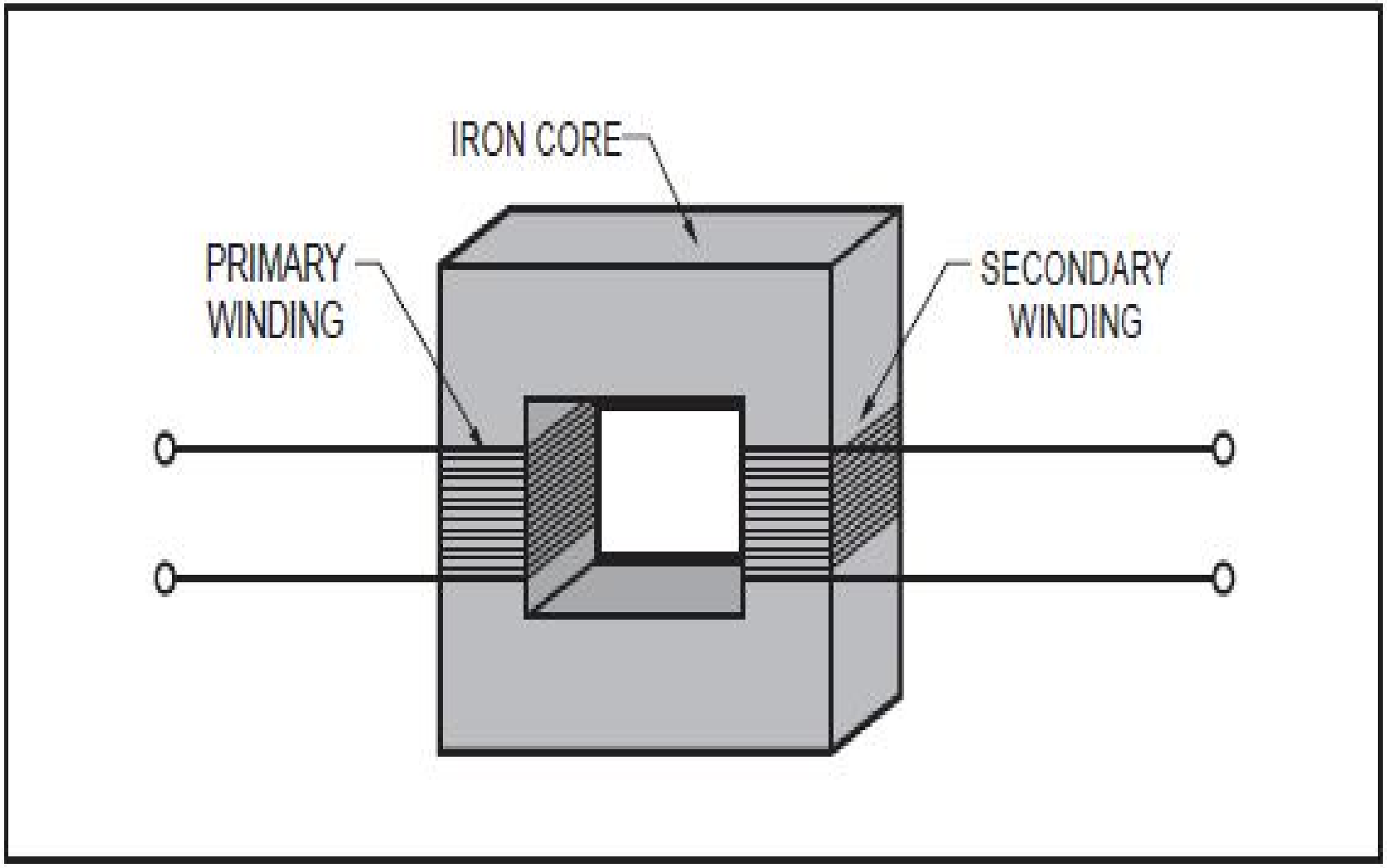




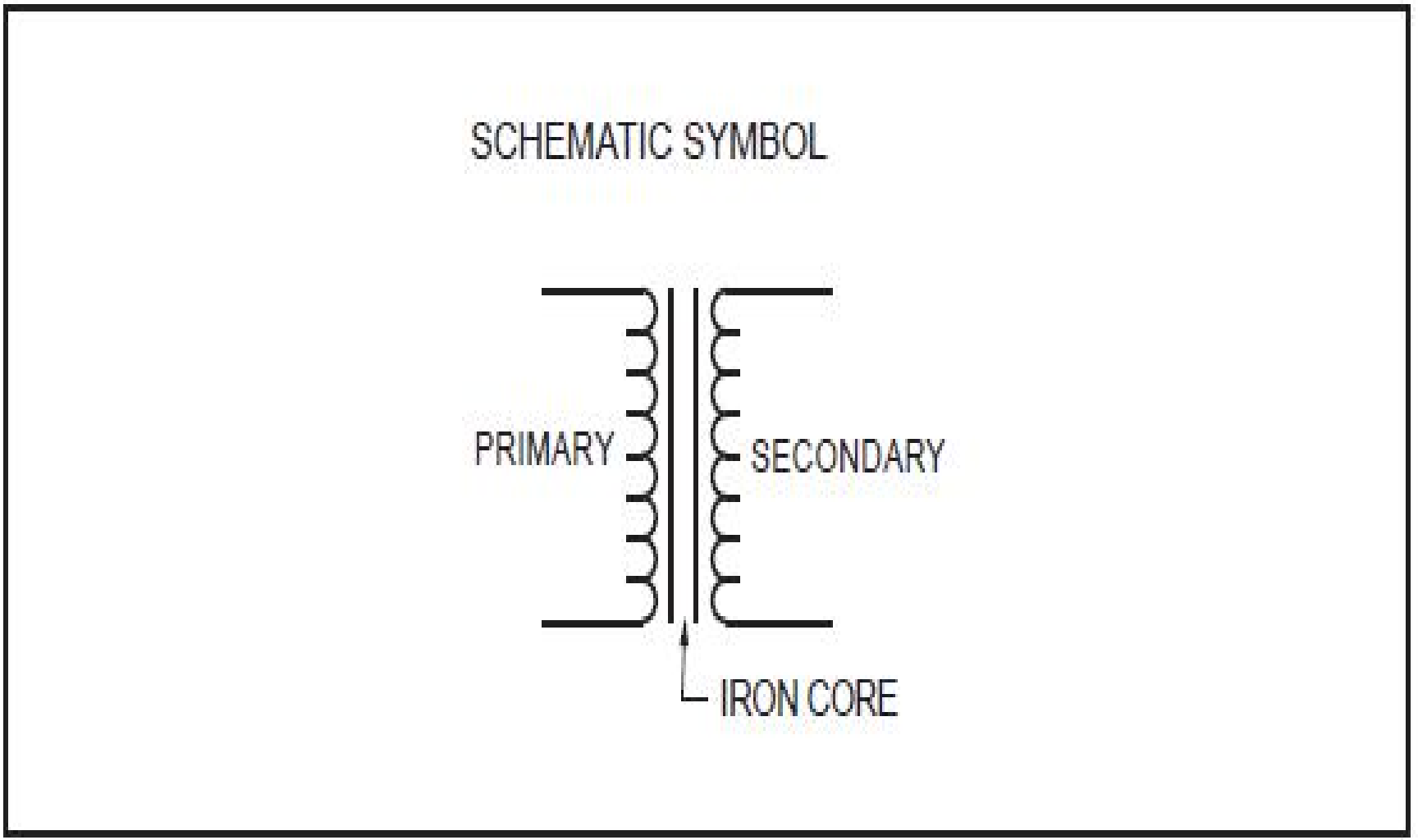


PART CROSS SECTION OF A VERY LARGE TRANSFORMER

Transformer Construction/structure



Symbol of Transformer

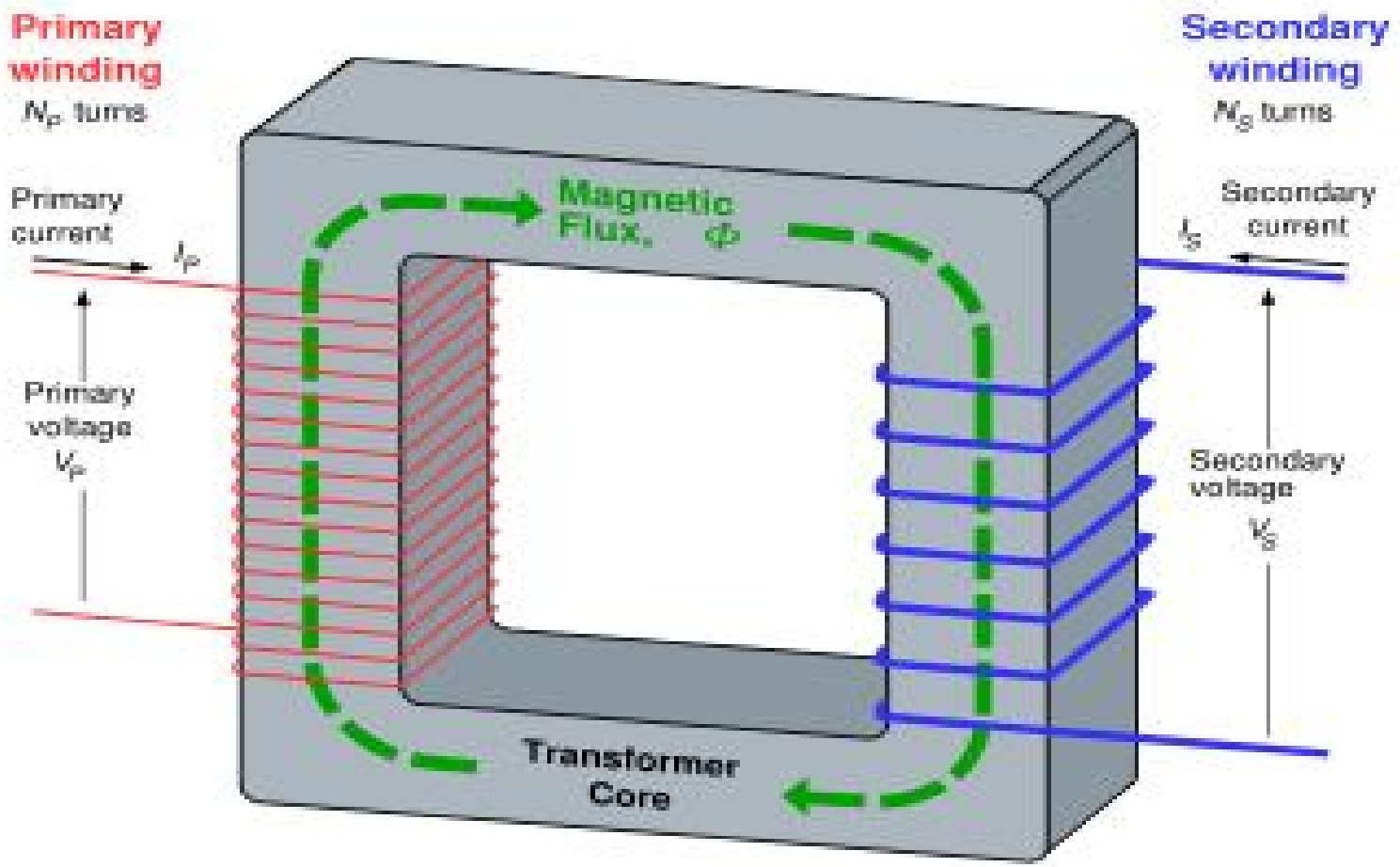


Parts of a Transformer

A transformer consists of 3 basic components

- Primary Coil or Primary Winding : It is an electrical wire wrapped around the core on the input side
- Secondary Coil or Secondary Winding: It is an electrical wire wrapped around the core on the output side
- Core : A ferromagnetic material that can conduct a magnetic field through it. Example: Iron

Transformer Structure



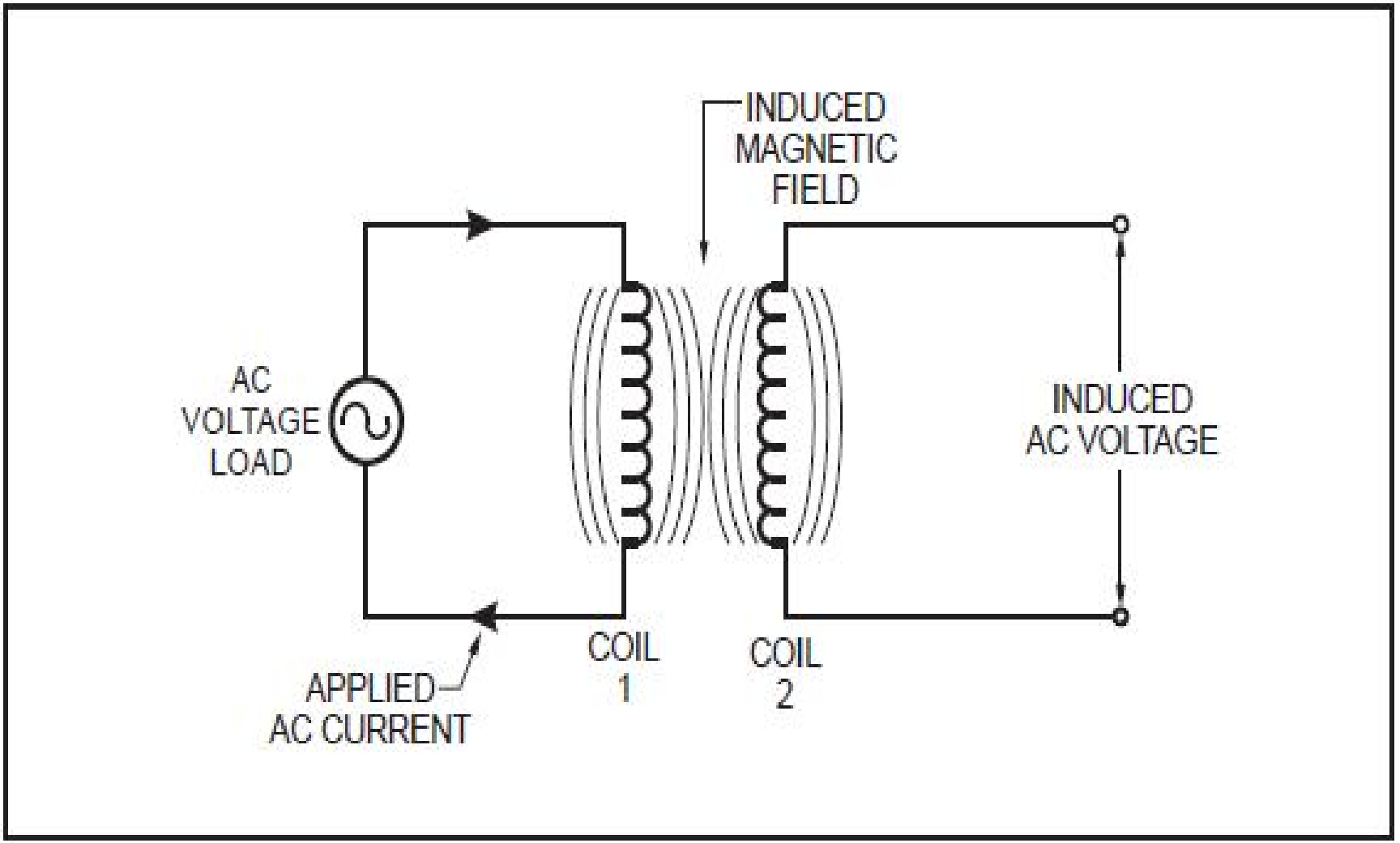
Transformer Operation

- An electrical transformer normally consists of a ferromagnetic core and two coils called "windings".
- A transformer uses the principle of mutual inductance to create an AC voltage in the secondary coil from the alternating electric current flowing through the primary coil.
- The voltage induced in the secondary can be used to drive a load.

What is Mutual Inductance?

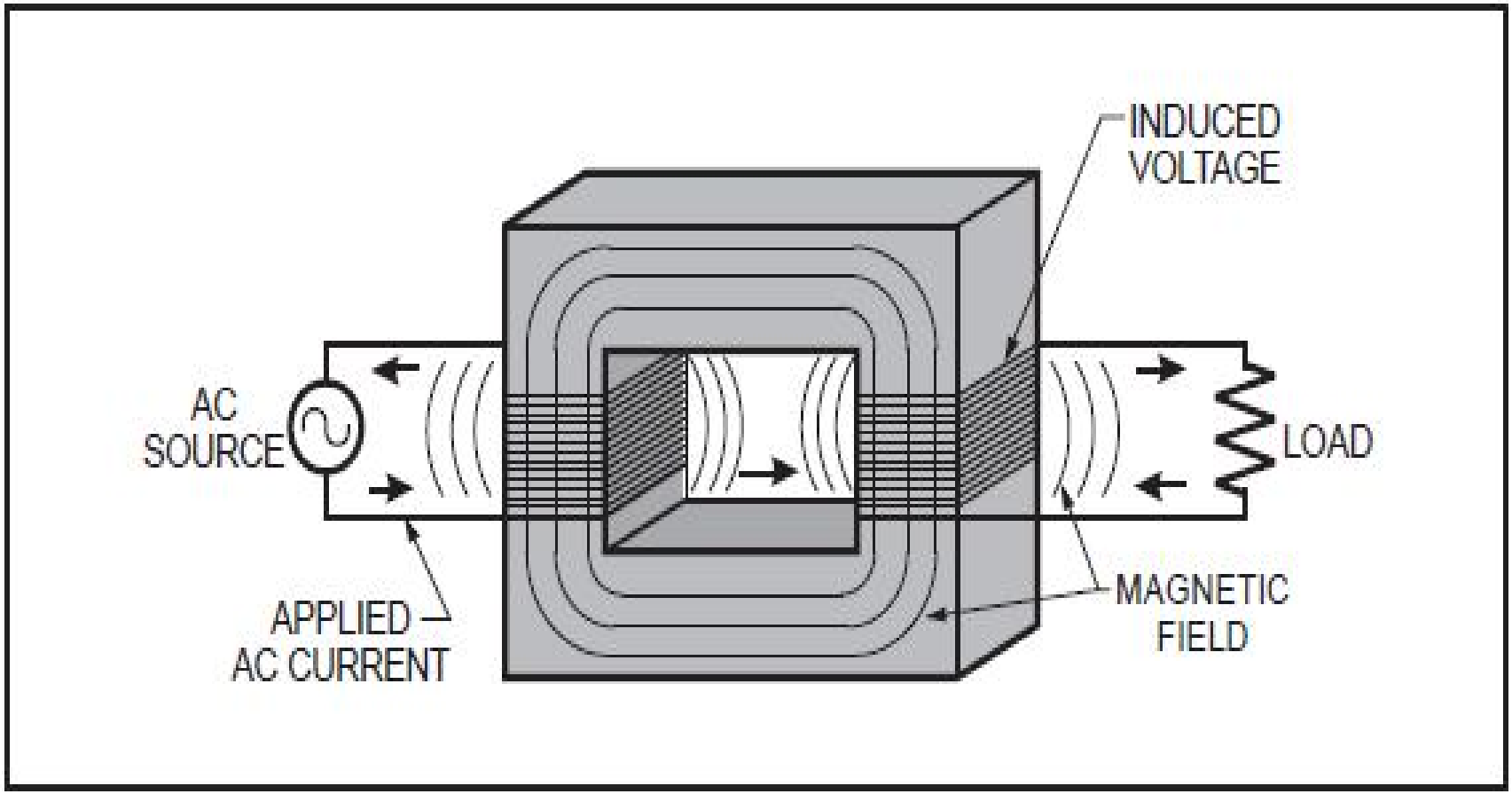
- The principle of mutual inductance says that when two electrical coils are placed near to each other, AC electrical current flowing in one coil induces an AC voltage in the other coil.
- This is because current in the first coil creates a magnetic field around the first coil which in turn induces a voltage in second coil

Mutual Inductance



- The transformer improves the efficiency of the transfer of energy from one coil to another by using a core to concentrate the magnetic field.
- The primary coil creates a magnetic field that is concentrated by the core and induces a voltage in the secondary coil

Transformer Operation



Turns Ratio

- The voltage at the secondary coil can be different from the voltage at the primary. This happens when the number of turns of the coil in primary and secondary are not the same
- The Turns Ratio (TR) is the ratio of the number of turns in the primary coil to the number of turns in the secondary coil

Transformer Formulas

TURNS RATIO FORMULA

$$TR = \frac{N_P}{N_S}$$

Where

N_p = *number of turns in the primary*

N_s = *number of turns in the secondary*

Continued…

TRANSFORMER OUTPUT VOLTAGE FORMULA

$$V_S = \frac{V_P}{TR}$$

Where

V_S = *secondary voltage (Volts)*
V_P = *primary voltage (Volts)*
TR = *turns ratio*

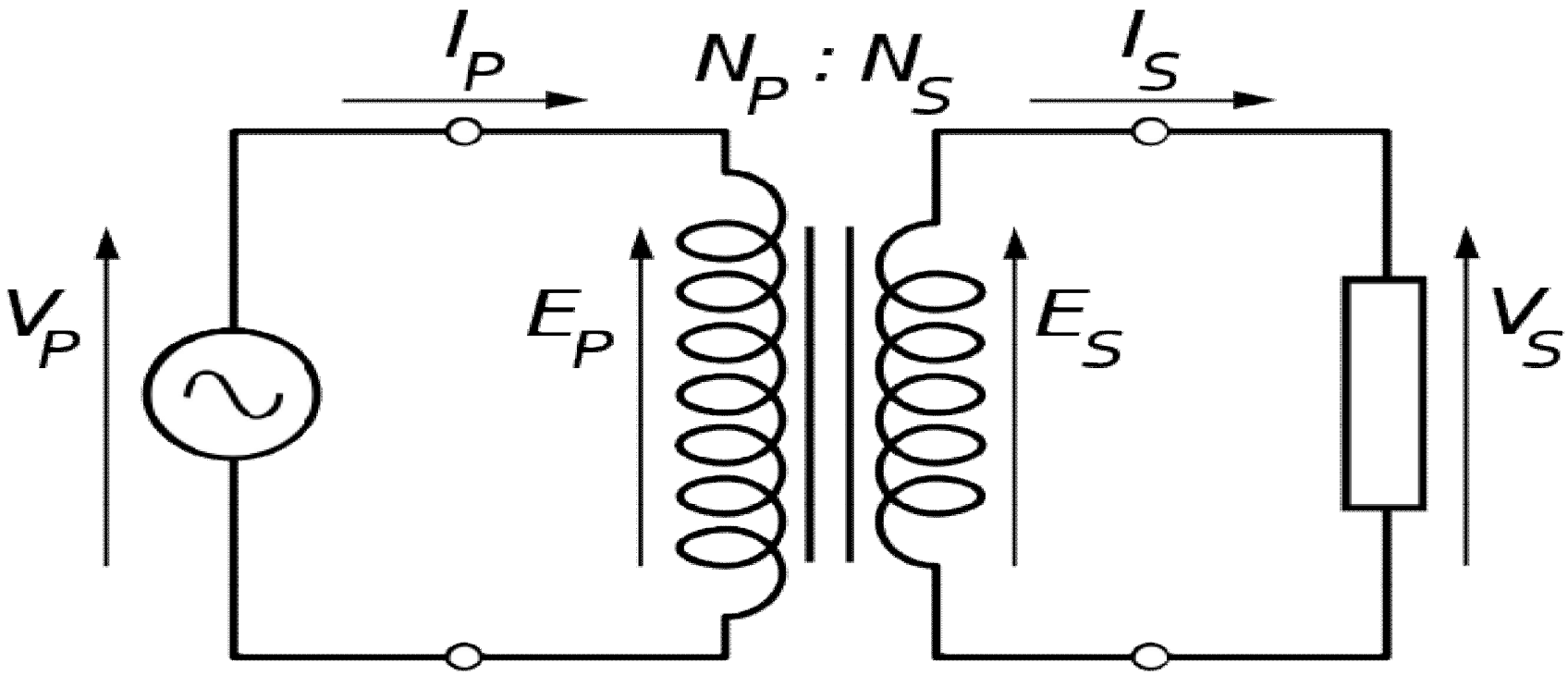
Formulas---

- TR= Vp / Vs
- Also TR= Np /Ns
- So we can say

Vp / Vs = Np /Ns

Also Vp/Vs=Is /Ip

Transformer Voltages & Currents



$$\frac{V_P}{V_S} = \frac{I_S}{I_P} = \frac{N_P}{N_S}$$

Problems

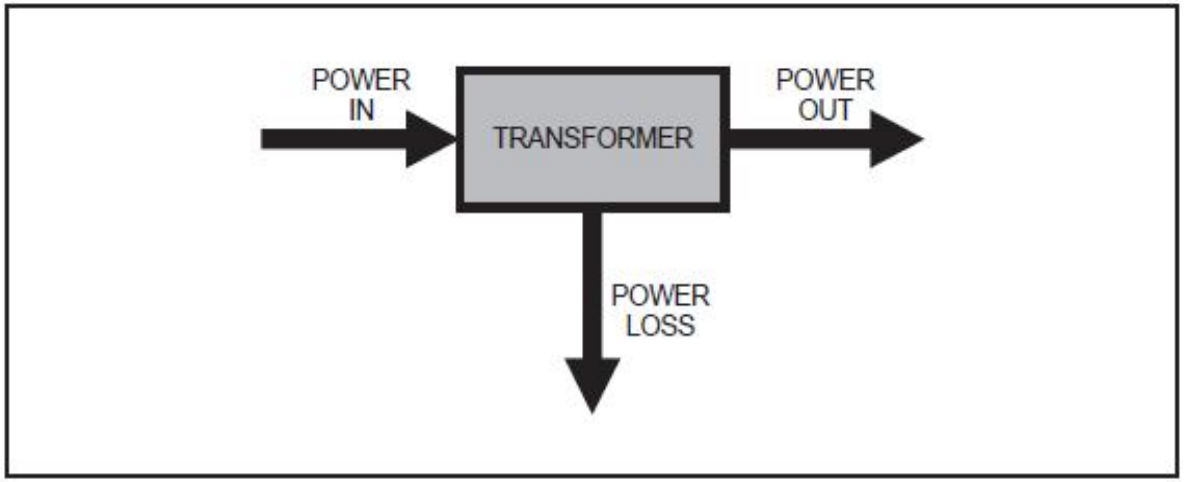
1. A transformer has a primary voltage of 230v and turns ratio of 5:1. Calculate the secondary voltage
2. A transformer has 200 turns in the primary, 50 turns in the secondary, and 120 volts applied to the primary (V_p). What is the voltage across the secondary (V_s)?

More Problems….

1. There are 400 turns of wire in an iron-core coil. If this coil is to be used as the primary of a transformer, how many turns must be wound on the coil to form the secondary winding of the transformer to have a secondary voltage of one volt if the primary voltage is five volts?
2. A 12 volts transformer has 20 turns in the primary, 5 turns in the secondary. What is the voltage across the primary side (V_P)?

Input Power and Output Power of a Transformer

- Under ideal conditions input power and output power should be the same. But there is power loss between the primary and secondary and so practically they are not exactly equal.
- So, $P_{in} = P_{out} + P_{loss}$



Transformer Efficiency

- The power loss is converted to heat . The heat produced can be found by calculating the transformer efficiency.

TRANSFORMER EFFICIENCY FORMULA

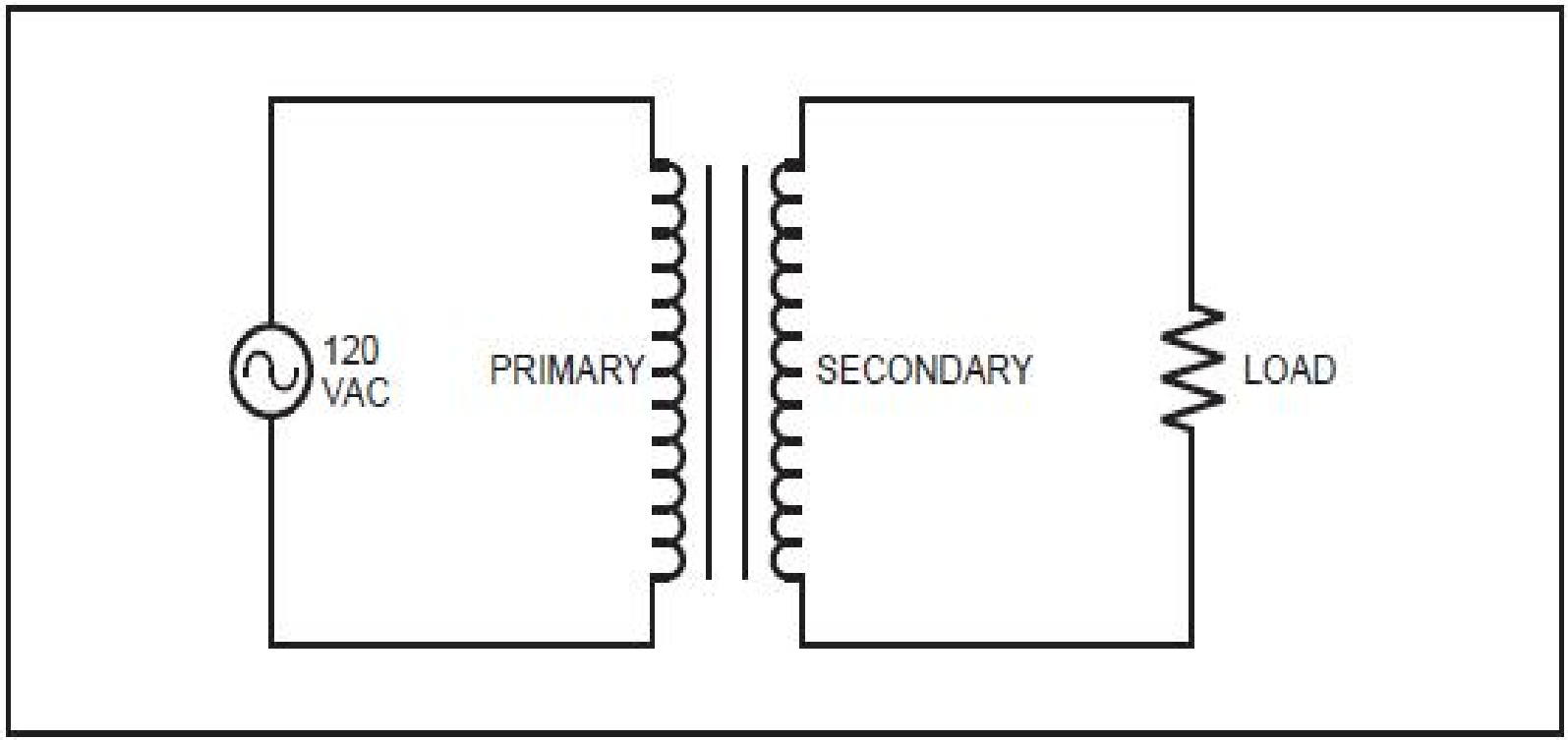
$$Transformer\ Efficiency\ \% = \frac{Power\ Out}{Power\ In} \times 100$$

Where

Power Out= output power (Watts or VA)
Power In = input power (Watts or VA)

Isolation Transformer

- In isolation transformer, the primary and secondary are physically isolated (no electrical connection)

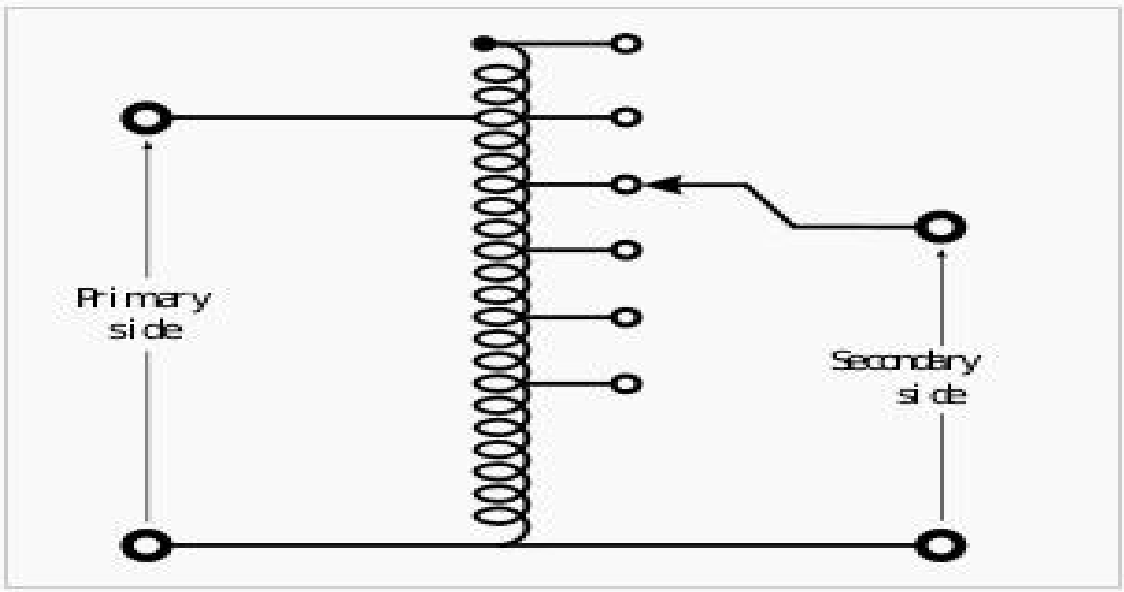


Advantages of Isolation Transformer

- Voltage spikes that might occur on the primary are greatly reduced or eliminated in the secondary
- If the primary is shorted somehow, any load connected to the secondary is not damaged
- Example: In TV monitors to protect the picture tube from voltage spikes in main power lines

Autotransformers

- An autotransformer uses only one coil for the primary and secondary.
- It uses taps on the coil to produce the different ratios and voltages.



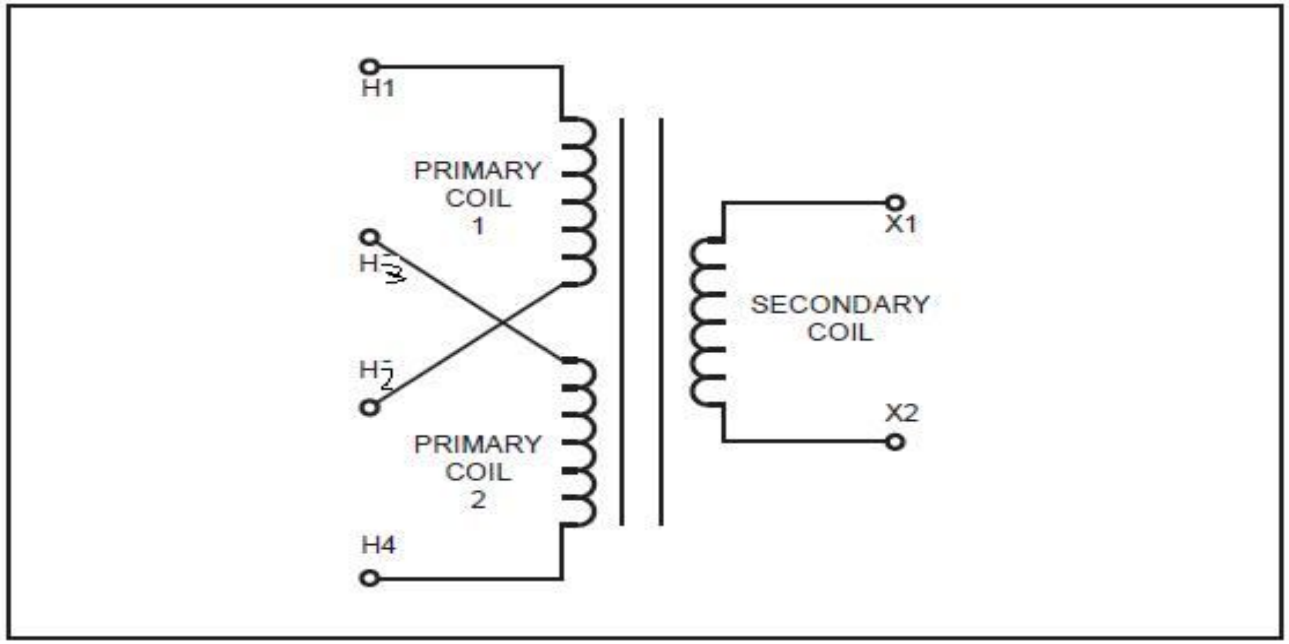
Single-phase tapped autotransformer with output voltage range of 40%—115% of input

The Control Transformer

- A control transformer is used to reduce voltage from the main power line to a lower voltage that operates a machine’s electrical control system.

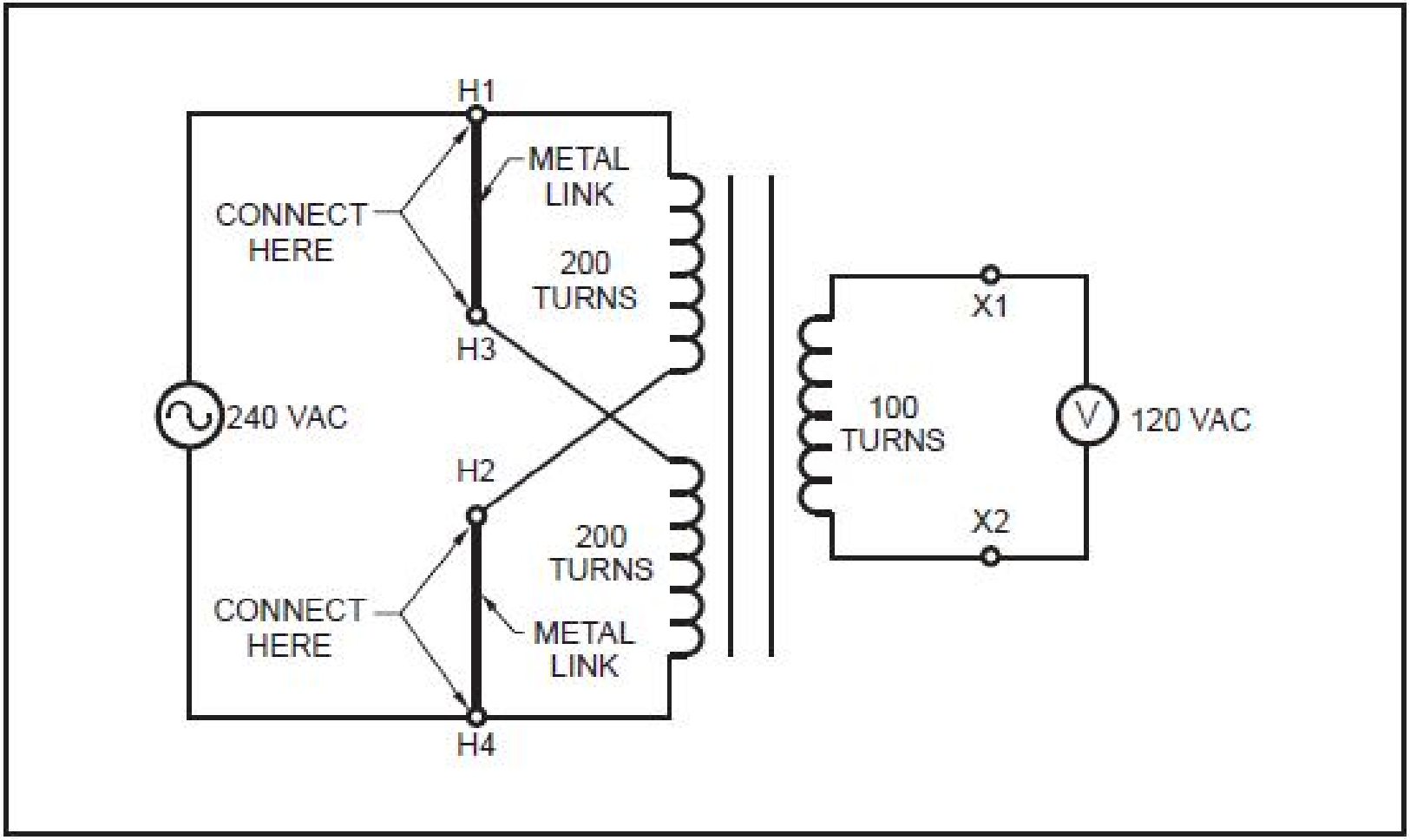
The Control Transformer

- The most common type of control transformer has two primary coils (H1H2 and H3H4) and one secondary coil (X1X2). Note that the primary windings are crossed



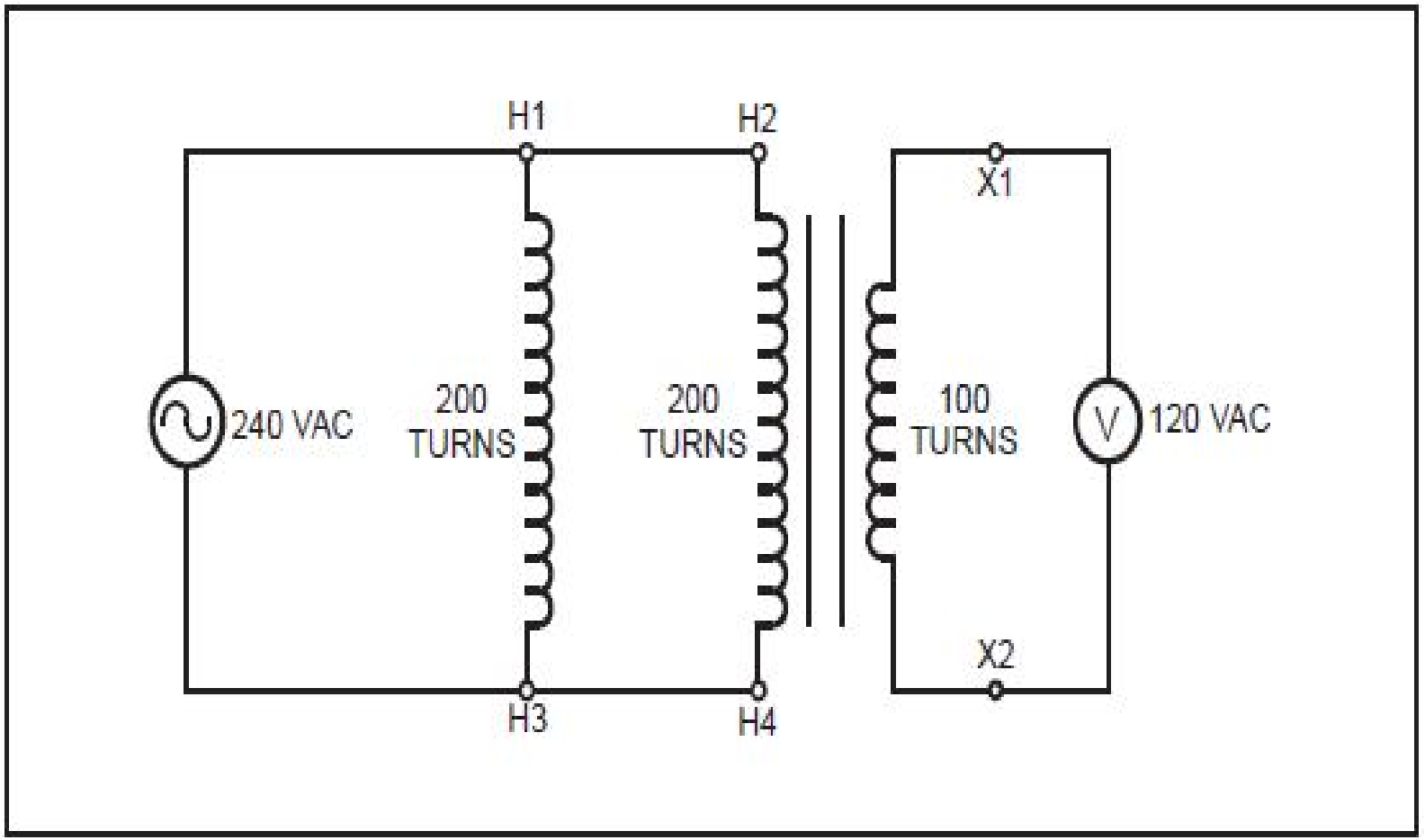
Continued…

To get 120V at the secondary from 240 V at the primary using a control transformer

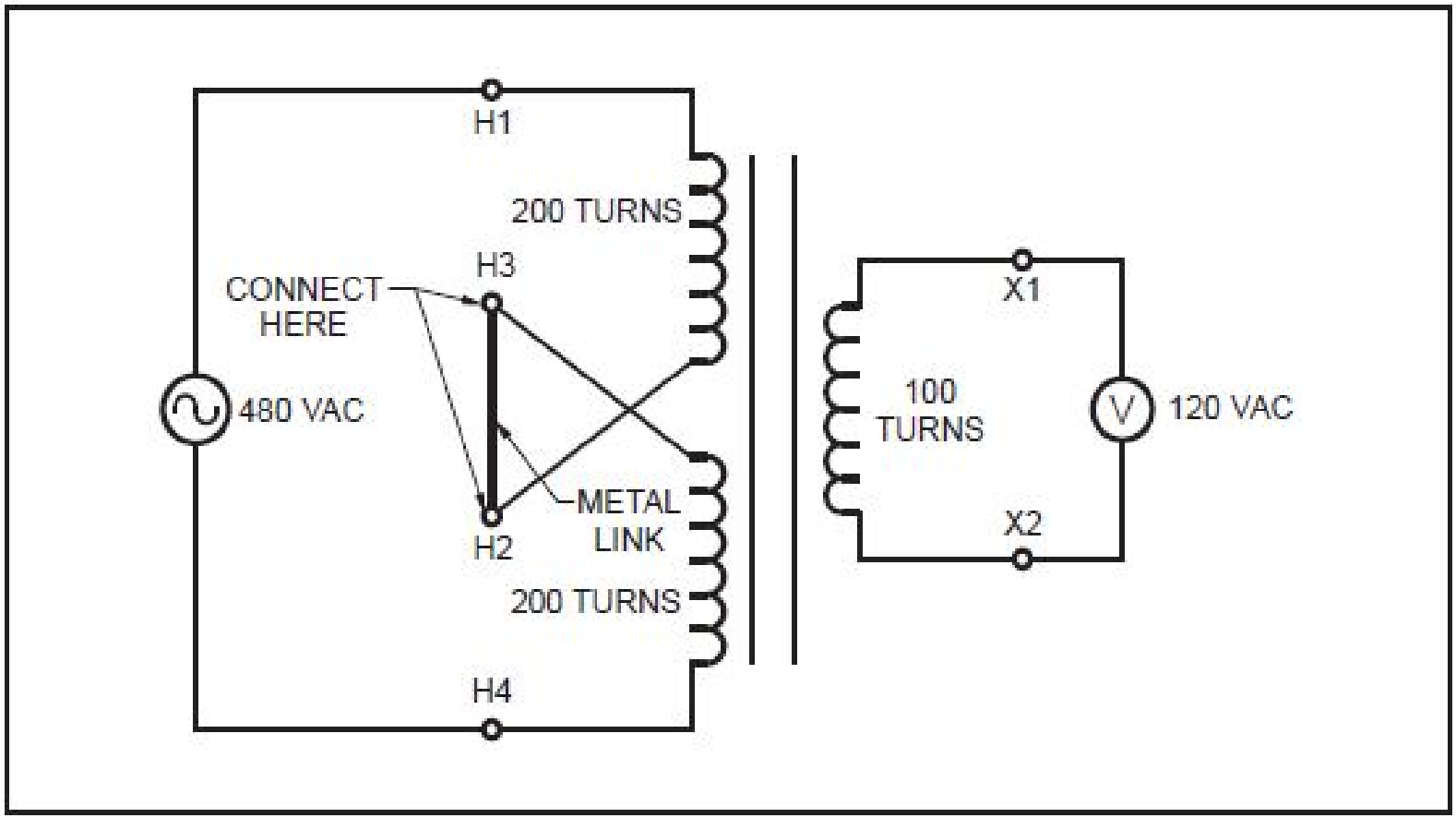


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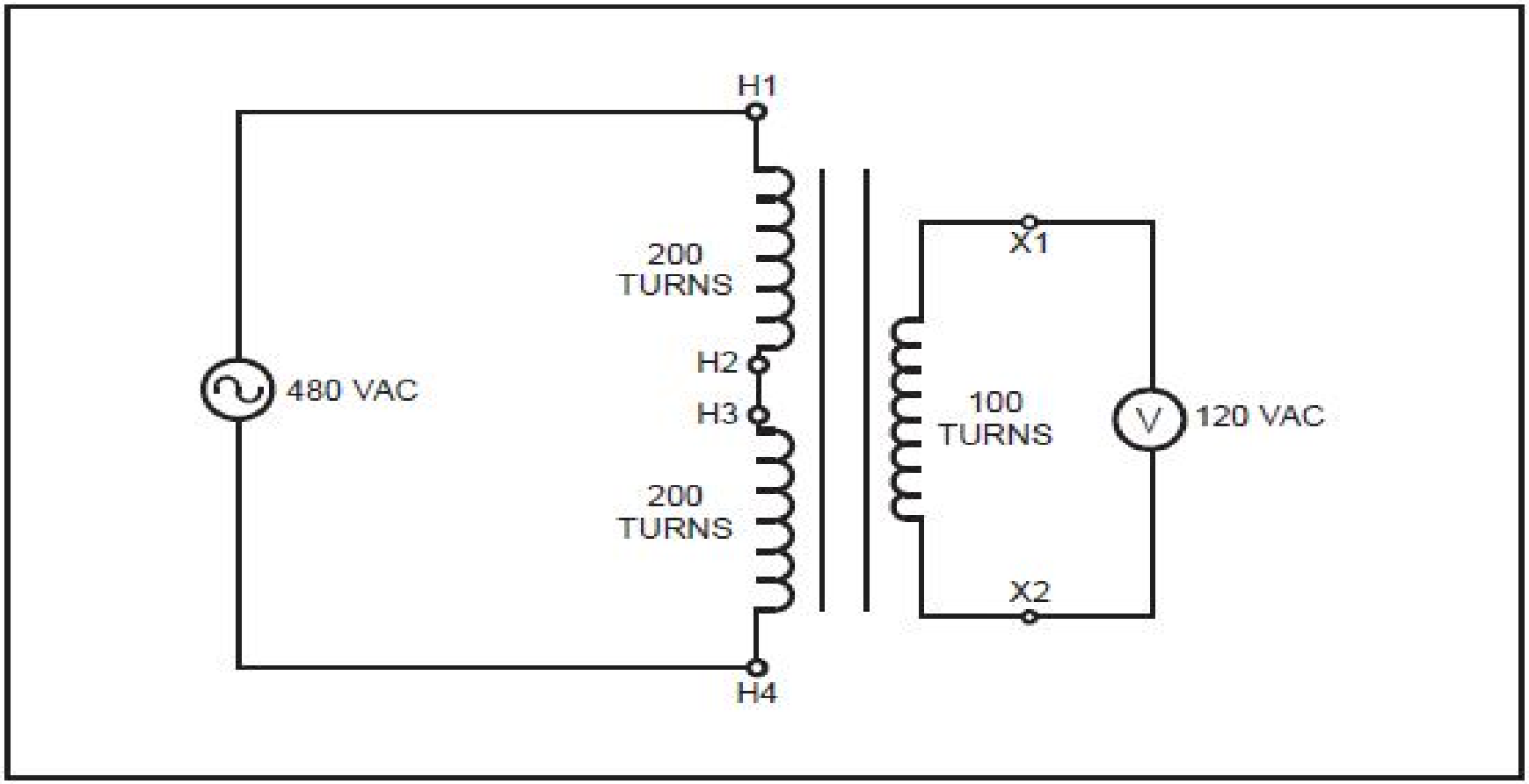
It is actually a parallel connection of the primary coils



To get 120V at the secondary from 480 V at the primary using a control transformer

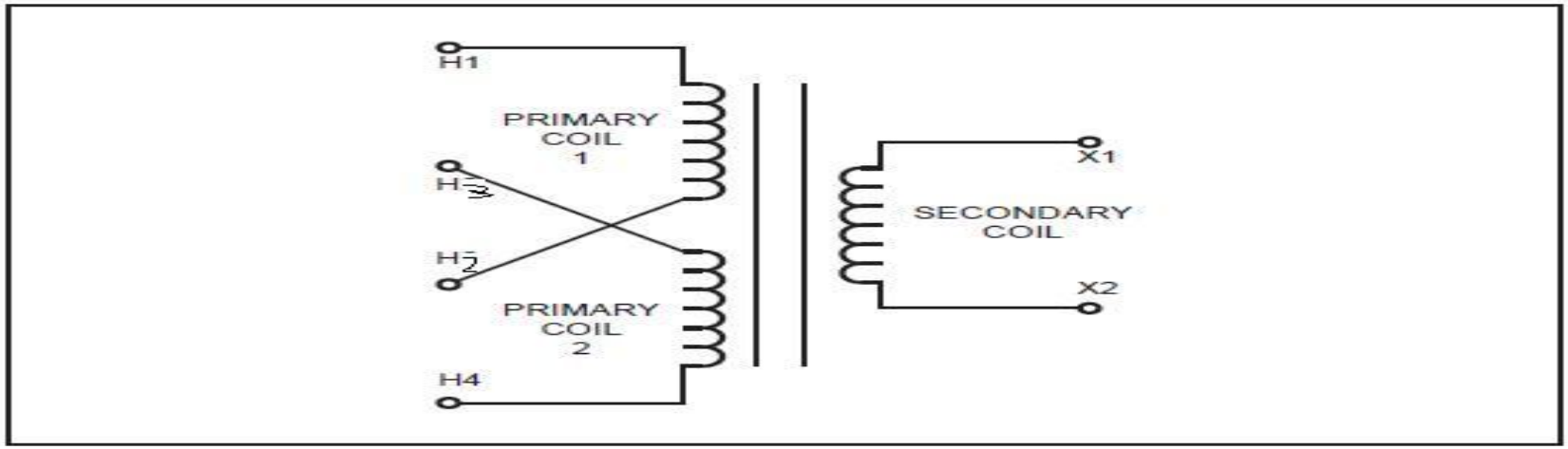


This is actually a series connection of the primary coils



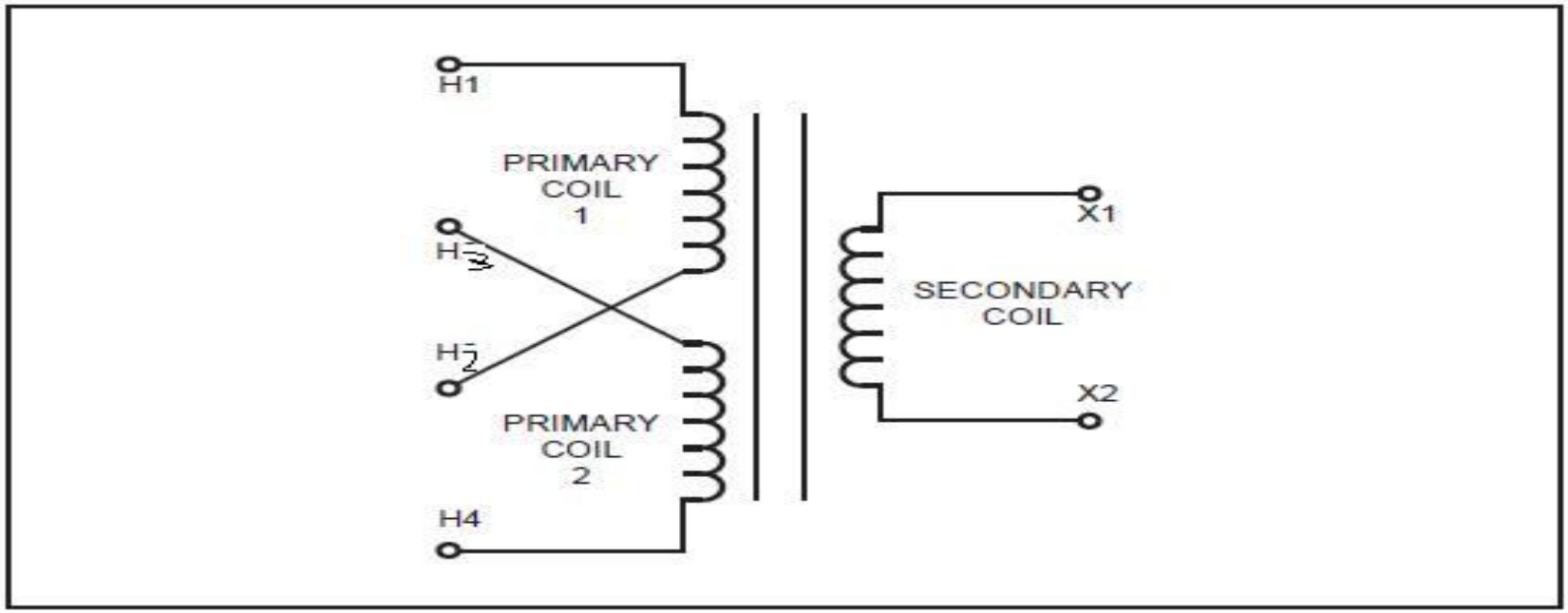
Problem 1

- Connect the primary coils in parallel and calculate the secondary voltage if the primary voltage is 48 Volts and the number of turns in each primary is 50 turns and the secondary has 25 turns.

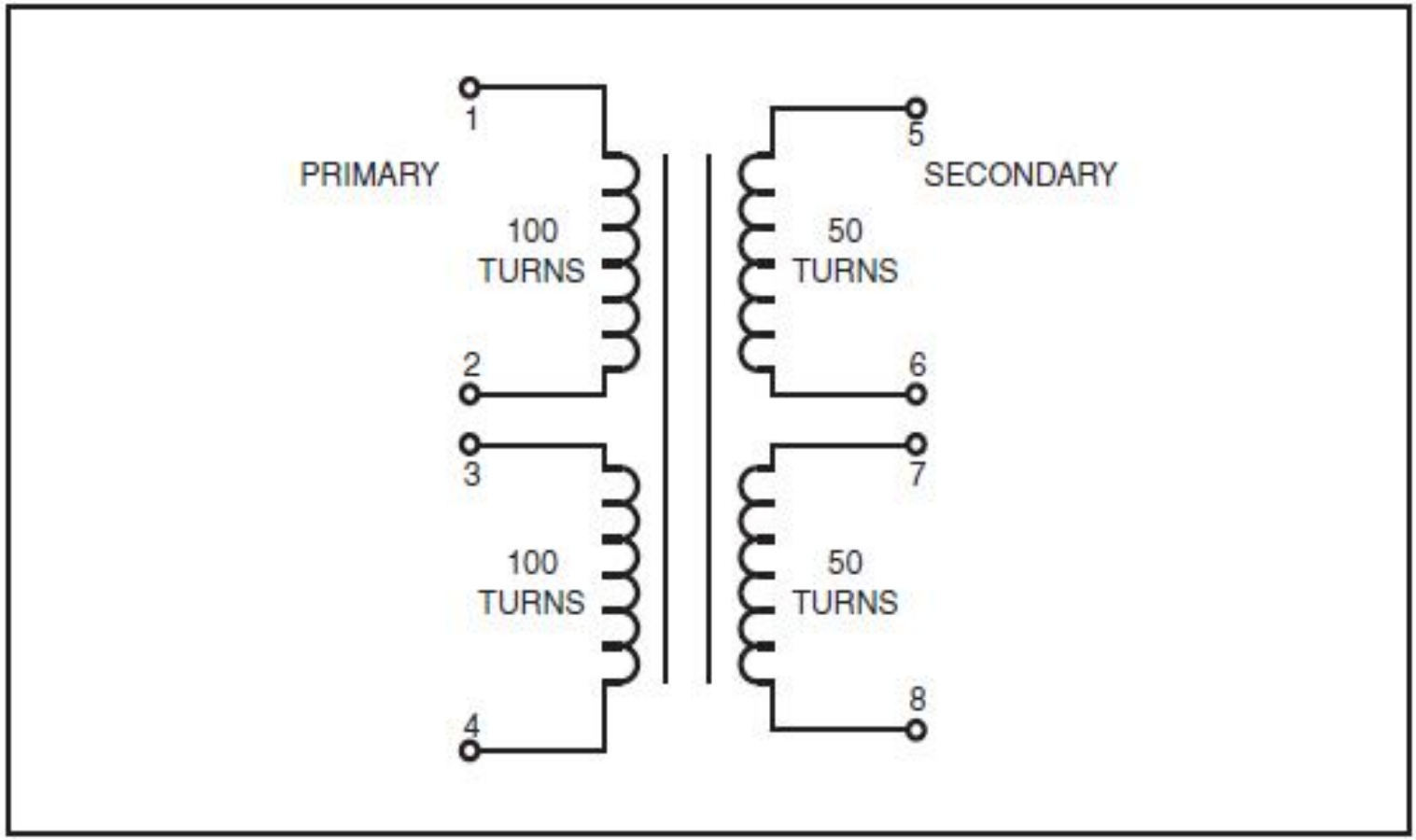


Problem 2

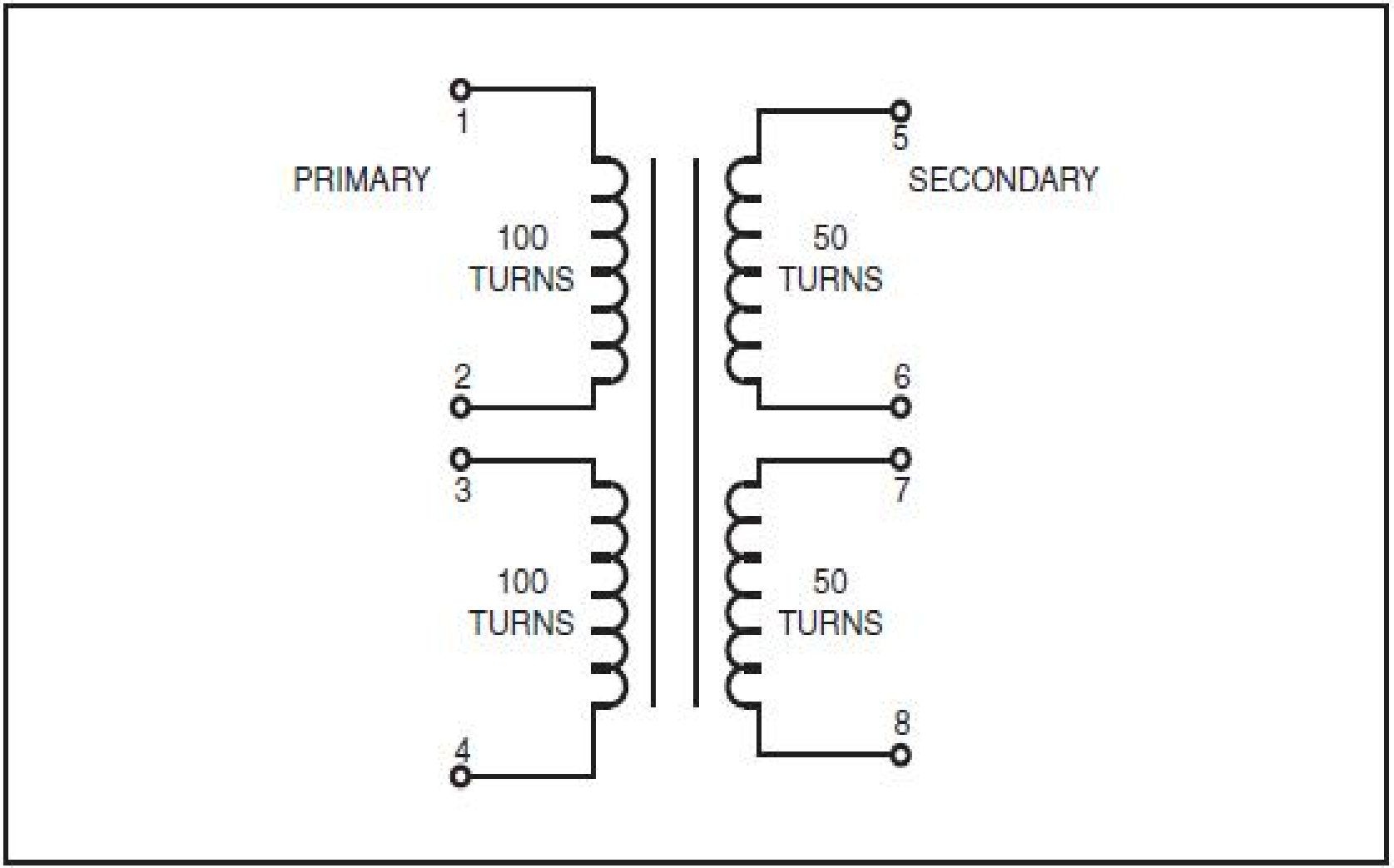
Connect the primary coils in series and calculate the secondary voltage if the primary voltage is 48 Volts and the number of turns in each primary is 50 turns and the secondary has 25 turns.



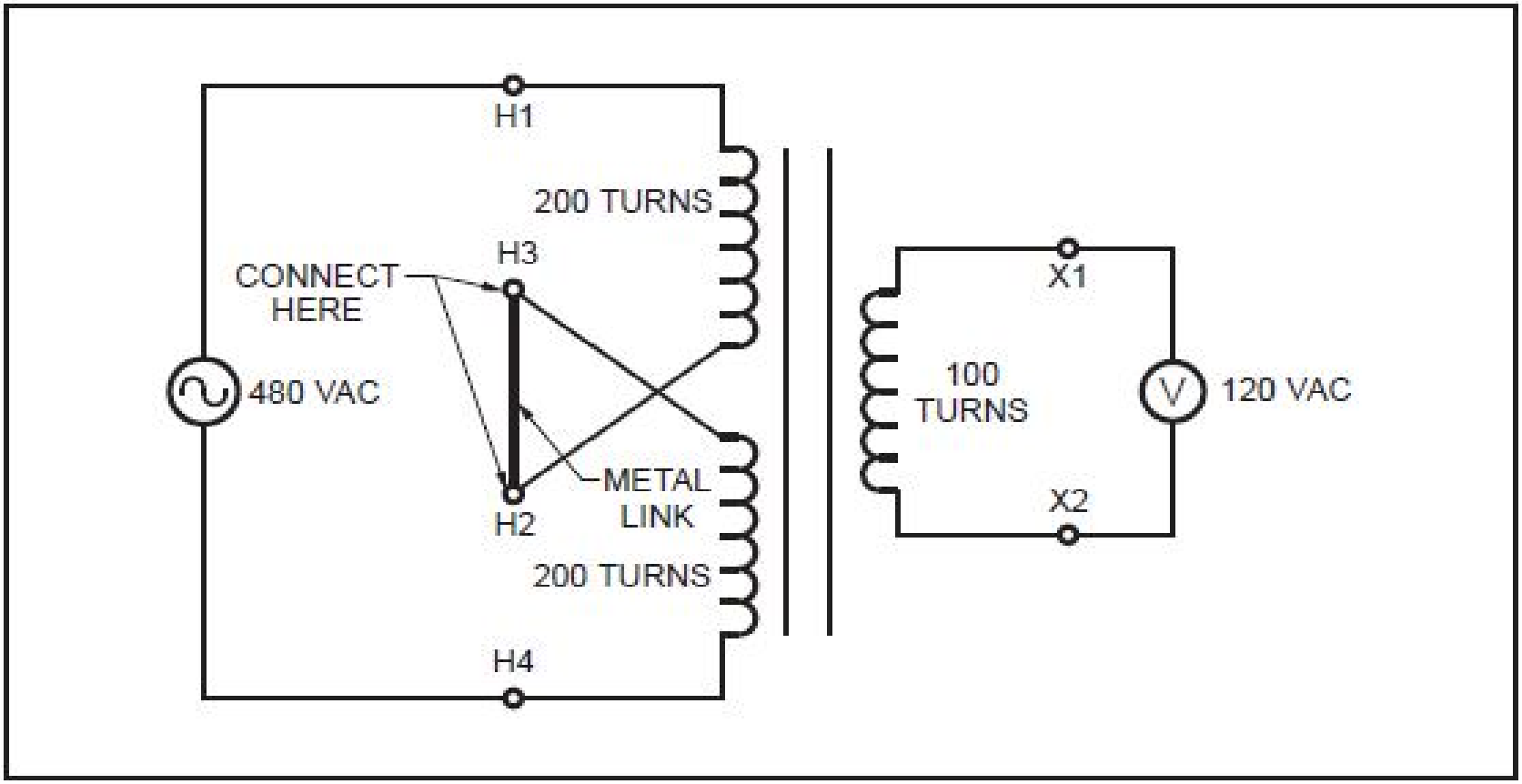
Problem 3: **Make connections in the transformer coils to produce a turns ratio of 1:1. Use both primary and secondary coils.**



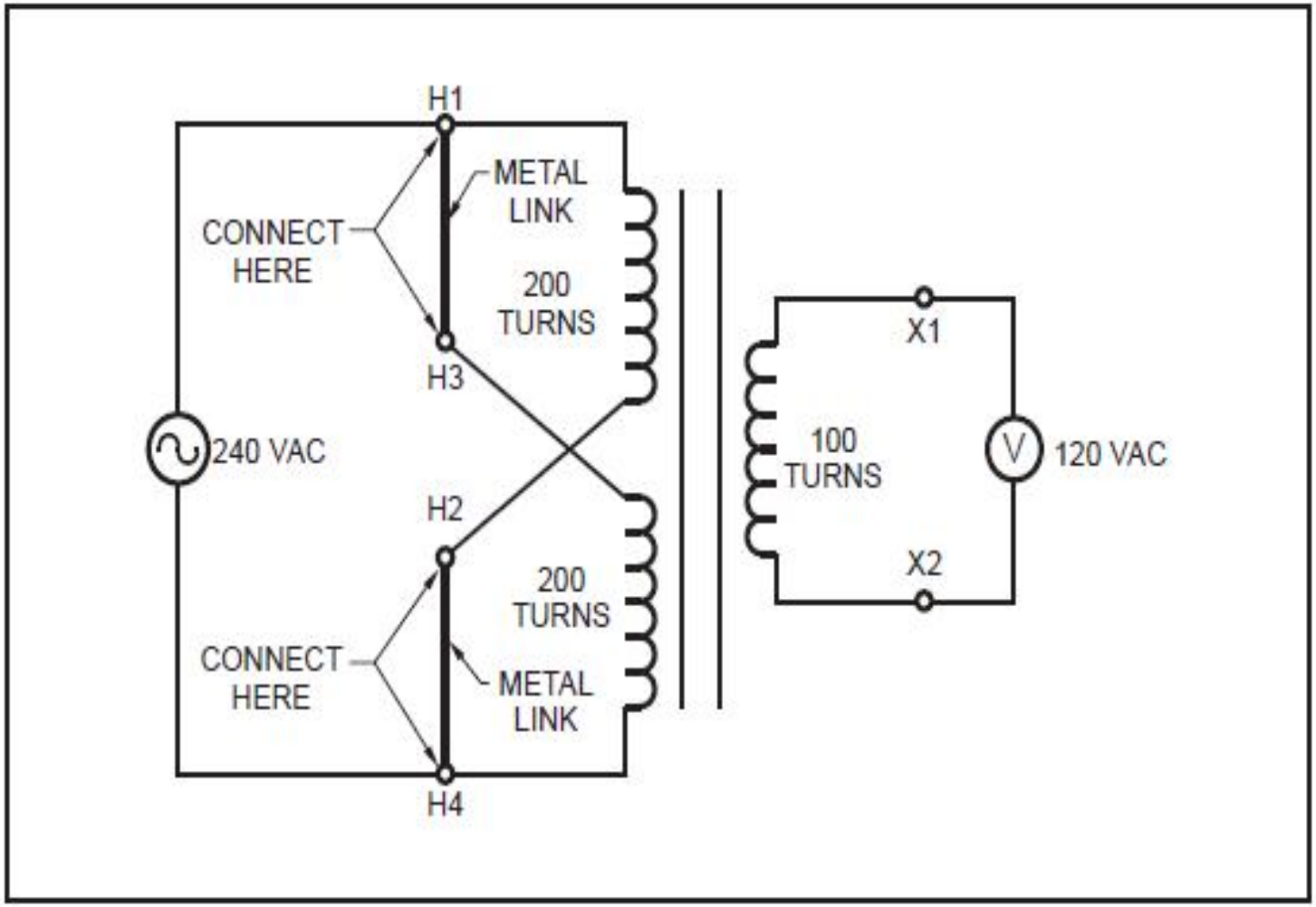
Problem 4: **Make connections in the transformer coils to produce a turns ratio of 2:1. Use both primary and secondary coils.**



Problem 5: What is the turns ratio in the circuit?



Problem 6: What is the turns ratio in this circuit?



Thank you for your attention!

I hope you will **“enjoy”** an **engineer’s life**!

Good Luck!!!